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## EXECUTIVE SUMMARY

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This Lean Six Sigma project was conducted to improve the packaging process for Ayrloom, a cannabis edibles division of Beak & Skiff. Ayrloom has been experiencing a steady 2–3% weekly growth in production, but its manual packaging process was not keeping pace. Workers were manually cleaning sticky gummies from cold storage, hand-packing 10 gummies per pouch, sealing the bags, and manually applying stickers. This highly repetitive, labor-intensive system had no formal work instructions or standard roles, which caused variation in worker performance and inconsistent output.

The team used the DMAIC (Define, Measure, Analyze, Improve, Control) framework. In the Measure phase, a time study was conducted across 9 workers with over 120 samples. The baseline showed that packaging productivity was about 47 pouches per person per hour, and there was a 4x variation in speed between fastest and slowest employees. The main bottleneck was cleaning which took as little as 20 seconds or as much as 80 seconds for the same task due to a lack of standardization and variation in gummy stickiness.

In the Analyze phase, tools such as Fishbone Diagrams, 5 Whys Analysis, and Spaghetti Diagrams were used to diagnose root causes. Major issues included inconsistent work methods, a disorganized floor layout, poor material flow, and excessive worker movement to fetch supplies.

The Improve phase focused on simple but high-impact changes. These included implementing a runner role to reduce movement waste, introducing task-based work allocation using Excel Solver, redesigning the layout to follow a linear workflow, and developing SOPs and visual aids.

A pilot test of these changes led to a 23.4% improvement in packaging productivity increasing from 47 to 58 pouches per person per hour. The cost per pouch dropped from \$0.43 to \$0.34, resulting in a projected annual labor savings of \$73,104 based on a post-pilot volume of 16,922 packets per week.

In the Control phase, the team implemented Excel-based time tracking, automated weekly role assignment using Solver, and a visual dashboard for monitoring productivity. Additional leadership and change management tools were introduced to ensure long-term sustainability.

Overall, this project not only improved packing efficiency but created a scalable, data-driven system that empowers frontline teams while delivering measurable financial value.

# DEFINE PHASE

## PROJECT CHARTER

|                                    |                                                                                                                                                                                                                                                |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Project Name</b>                | Improving Gummy Packing Efficiency at Ayrloom                                                                                                                                                                                                  |
| <b>Business/Location</b>           | Beak & Skiff (Ayrloom Division), 4472 Cherry Valley Trpk, Route 20                                                                                                                                                                             |
| <b>Team Leader</b>                 | Meet Patel                                                                                                                                                                                                                                     |
| <b>Champion</b>                    | Bryanna Cortese, COO                                                                                                                                                                                                                           |
| <b>Project Description/Mission</b> | Enhance productivity in the gummy filling operation at Ayrloom by optimizing the manual packing process. The goal is to increase the packing efficiency from 50-60 pouches per hour to 70 pouches per hour per person.                         |
| <b>Problem Statement</b>           | The company's gummy production volume is increasing by 2-3% per week. The current packaging methods produce 50-60 pouches per hour per person, which does not match the growing production rate.                                               |
| <b>Business Case</b>               | Enhancing the packing process will lead to increased productivity and cost savings. This will improve operational efficiency, support business growth in the expanding cannabis market and maintain product quality and customer satisfaction. |
| <b>Deliverables</b>                | <ul style="list-style-type: none"> <li>Improved packing process flow</li> <li>KPIs for performance tracking</li> <li>Training materials for staff on new procedures</li> </ul>                                                                 |
| <b>Goals/Metrics</b>               | <ul style="list-style-type: none"> <li>Increase packing efficiency to 70 packets/hour/ person</li> <li>Establish KPIs to monitor ongoing performance</li> </ul>                                                                                |
| <b>Process &amp; Owner</b>         | Packing Process Owner: Chelsea Bugsch                                                                                                                                                                                                          |
| <b>Project Scope Is</b>            | <ul style="list-style-type: none"> <li>Optimization of the manual packing process</li> <li>Implementation of error-proofing measures</li> <li>Training and development of packing staff</li> </ul>                                             |

|                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Project Scope Is Not</b>       | <ul style="list-style-type: none"> <li>• Redesigning the entire supply chain</li> <li>• Changing external suppliers</li> <li>• Modifying product formulations</li> </ul>                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Key Customers</b>              | <ul style="list-style-type: none"> <li>• Internal: Packing Team, Quality Control Team</li> <li>• External: Retailers and End Consumers</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Customer Expectations</b>      | <ul style="list-style-type: none"> <li>• Consistent product quality</li> <li>• Timely delivery of products</li> <li>• Reduction in packaging errors</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Milestones</b>                 | <ul style="list-style-type: none"> <li>• Project Start: 01/31/25</li> <li>• Define Phase: 01/31/25</li> <li>• Measure Phase: 02/21/25</li> <li>• Analyze Phase: 03/11/25</li> <li>• Improve Phase: 03/31/25</li> <li>• Control Phase: 04/17/25</li> </ul>                                                                                                                                                                                                                                                                                             |
| <b>Expected Business Benefits</b> | <ul style="list-style-type: none"> <li>• <b>Soft Cost Savings:</b> Lower rework costs</li> <li>• <b>Speed:</b> Faster packing times</li> <li>• <b>Compliance:</b> Enhanced quality control and audit processes</li> <li>• <b>Intangible:</b> Improved team morale and customer satisfaction</li> </ul>                                                                                                                                                                                                                                                |
| <b>Est. Savings</b>               | <p>By improving the packaging efficiency from 50 packets per hour to 70 packets per hour, the cost per bag is reduced from \$0.40 to \$0.29. This efficiency gain results in a savings of \$0.11 per bag. Based on a weekly production volume of 9,692 bags, the projected cost savings are as follows:</p> <ul style="list-style-type: none"> <li>• <b>Weekly Savings:</b> <math>9,692 \times \\$0.11 = \\$1,066</math></li> <li>• <b>Monthly Savings:</b> Approximately \$4,264</li> <li>• <b>Annual Savings:</b> Approximately \$51,173</li> </ul> |

|                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Team Members</b>                 | <ul style="list-style-type: none"> <li>• Sai Santosh Gangawane – Engineering Management</li> <li>• Katherine Rodrigues Acosta – MBA</li> <li>• Meet Patel – Engineering Management</li> <li>• Meghanath Somarowthu – Business Analytics</li> <li>• Ashish Kushwaha – Data Science</li> <li>• Carrie - Packing Lead</li> <li>• Ana - Packaging Team</li> <li>• Evan - Packaging Team</li> <li>• Tessa - Packaging Team</li> <li>• Lisa - Packaging Team</li> <li>• Madaline - Packaging Team</li> <li>• Morgan - Packaging Team</li> </ul> |
| <b>Expected Resource Needs</b>      | Data collection tools for performance tracking                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Risk Assessment</b>              | <ul style="list-style-type: none"> <li>• Resistance to change from staff</li> <li>• Potential equipment malfunctions</li> <li>• Supply chain delays impacting implementation</li> </ul>                                                                                                                                                                                                                                                                                                                                                   |
| <b>Roles &amp; Responsibilities</b> | <ul style="list-style-type: none"> <li>• <b>Carrie (Packing Lead):</b> Oversees the packing process, manages daily operations, and facilitates training.</li> <li>• <b>Packaging Team (Ana, Evan, Tessa, Lisa, Madaline, Morgan):</b> Responsible for filling pouches, sealing, conducting weight checks, and reporting discrepancies</li> </ul>                                                                                                                                                                                          |
| <b>Prepared By</b>                  | <ul style="list-style-type: none"> <li>• Meet Patel – Engineering Management</li> <li>• Sai Santosh Gangawane – Engineering Management</li> <li>• Meghanath Somarowthu – Business Analytics</li> <li>• Katherine Rodrigues Acosta – MBA</li> <li>• Ashish Kushwaha – Data Science</li> </ul>                                                                                                                                                                                                                                              |
| <b>Date (Last Revision)</b>         | Feb 15, 2025                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

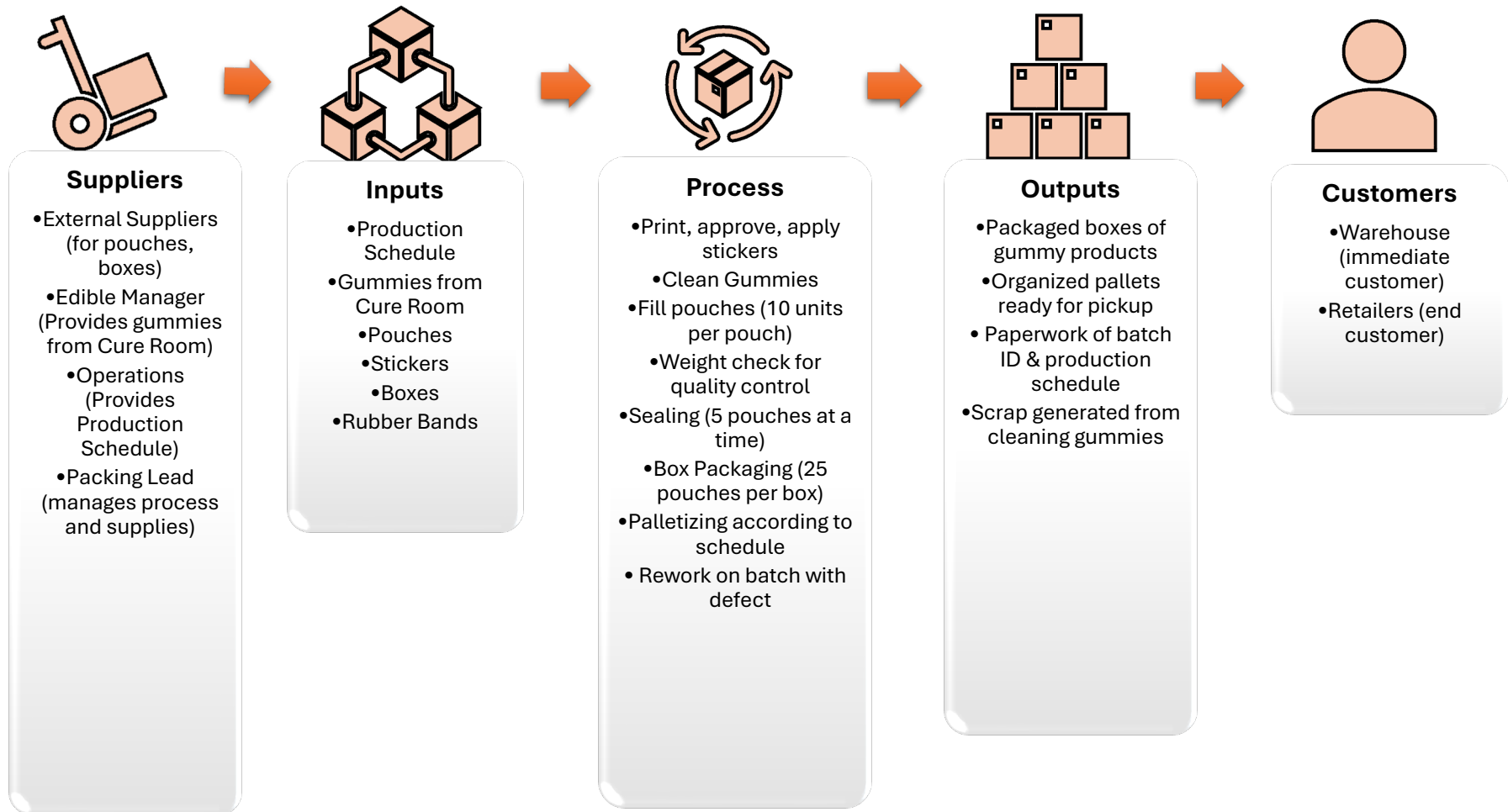
## COMMUNICATION PLAN

| <b>Audience</b>     | <b>Media</b>            | <b>Purpose</b>                         | <b>Key Messages</b>                                        | <b>Owner</b> | <b>Frequency</b> | <b>Notes</b>                                                                           |
|---------------------|-------------------------|----------------------------------------|------------------------------------------------------------|--------------|------------------|----------------------------------------------------------------------------------------|
| <b>Bryanna</b>      | Weekly Emails/ Meetings | Track progress and ensure alignment    | Progress on the process of solving the problems & feedback | Lean Team    | Weekly           | Visits to be planned and should cover specific objectives to ensure minor disruptions. |
| <b>Carrie (TL)</b>  | Weekly Meetings         | Tracking the Challenges                | Process changes and feedback                               | Brianna      | Weekly           |                                                                                        |
| <b>Line Workers</b> | Training Sessions       | Explain changes and encourage adoption | New processes                                              | Carrie       | Whenever needed  | Provide training                                                                       |

## CRITICAL TO QUALITY

| <b>Customer</b>      | <b>Need</b>               | <b>Driver</b>                            | <b>CTQC</b>                                                                                     |
|----------------------|---------------------------|------------------------------------------|-------------------------------------------------------------------------------------------------|
| <b>End customers</b> | Package of gummies        | Availability of product                  | <ul style="list-style-type: none"> <li>Stockout rate</li> </ul>                                 |
|                      | Value for money           | True to quantity as mentioned on packets | <ul style="list-style-type: none"> <li>Item count defect rate</li> </ul>                        |
|                      | Fresh gummies (not soggy) | Good quality product                     | <ul style="list-style-type: none"> <li>Best Use by date</li> <li>Customer complaints</li> </ul> |

## SIPOC ANALYSIS



# MEASURE PHASE

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## DATA COLLECTION PLAN

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A structured Data Collection Plan was developed to guide the Measure phase of the project. The purpose of this plan is to ensure that all key process metrics are monitored systematically, using accurate and consistent data sources. The KPIs selected reflect performance across critical stages of the gummy packaging process: cleaning, packing, and overall manufacturing cost and efficiency.

Each performance measure was chosen to address a specific aspect of the process that directly influences productivity or cost. Packaging Productivity provides a direct view of throughput per worker. Cleaning and Packing Time per Bin measure efficiency and reveal potential bottlenecks. Production Cost per Packet helps evaluate cost-effectiveness at the unit level and understand the financial impact of operational inefficiencies. Collecting these metrics allowed the team to diagnose root causes, identify variability, and lay the foundation for data-driven improvements.

In addition to standard data sources, a dedicated time study was conducted as part of the data collection effort. The team observed 9 workers and collected approximately 120 samples to track the time spent cleaning and packing 10 gummies per batch. Multiple readings were taken for each worker to capture variability in performance across shifts. The time data was used to calculate average task duration and standard deviation, enabling a clearer understanding of productivity baselines and individual work pace. This time study provided essential insights that guided subsequent analysis and improvement decisions.

| <b>Performance Measure</b>                        | <b>Data Source</b>                       | <b>How will Data be collected</b>                                           | <b>Who will collect Data</b>       | <b>When will Data be collected?</b>  | <b>Sample Size</b>    |
|---------------------------------------------------|------------------------------------------|-----------------------------------------------------------------------------|------------------------------------|--------------------------------------|-----------------------|
| <b>Packaging Productivity (Units/hour/person)</b> | Daily Production Logs, Payroll Sheet     | Count total packed units & divide by total hours worked and no. Of workers. | Manufacturing Head                 | End of each week                     | All workers per shift |
| <b>Cleaning Time per Bin (Minutes/bin)</b>        | Cleaning Station Log Sheet               | Timestamp start & finish time for each bin cleaned                          | Workers                            | During shift, each cleaning cycle    | 4-6 Records per Week  |
| <b>Packing Time per Bin (Minutes/bin)</b>         | Packing Station Sheets                   | Timestamp start & end time for packing each bin                             | Workers                            | During shift, each packing cycle     | 4-6 Records per Week  |
| <b>Packing Cost per Packet (\$)</b>               | Payroll sheet, Production log            | Pay per hour divide by the units made per person per hour                   | Manufacturing Head                 | Weekly or biweekly                   | 1 Record per Week     |
| <b>Time to Clean 10 Gummies (Minutes)</b>         | Time Study Sheet - Cleaning Area         | Timestamp start & end of each 10-gummy cleaning cycle                       | Meghanath, Meet, Katherine, Ashish | During shift, each cleaning instance | 60-100 records        |
| <b>Time to Pack 10 Gummies (Minutes)</b>          | Time Study Sheet - Packing Area          | Timestamp start & end of each 10-gummy packing cycle                        | Ashish, Katherine, Meet, Sai       | During shift, each packing instance  | 60-100 records        |
| <b>Time to Assemble Labeled Boxes (Minutes)</b>   | Manual Observation Log - Boxing Station  | Time measurement of box folding and labeling                                | Sai                                | As needed during packaging rounds    | 15-20 samples         |
| <b>Time to Sticker Packets (Minutes)</b>          | Manual Observation Log - Sticker Station | Measure time to sticker a full packet set (10 units)                        | Sai                                | During shift, post-packing cycle     | 15-20 samples         |

## TIME STUDY

As part of the Measure phase, a focused time study was conducted to evaluate how long workers took to complete two critical tasks in the gummy packaging process: cleaning and packing. Specifically, we observed and recorded the time required to clean 10 gummies and to pack 10 gummies into sealed packets. Each observation captured the start and end times of the task using manual time logs and stopwatches. The study was carried out during regular production shifts, and data was collected by a team of observers including Meet, Katherine, Meghanath, Ashish, and Sai.

Two separate sheets were maintained—one for cleaning and one for packing—with multiple observations recorded per worker to capture variations in performance. We collected approximately 120 samples for cleaning and 120 samples for packing, covering 9 different workers over multiple days. Figure 1 and Figure 2 below show a sample from the actual time study logs from the floor during the data collection period.

|    | A                                                                                          | C    | D     | E    | F        | G      | H    | I       | J   |
|----|--------------------------------------------------------------------------------------------|------|-------|------|----------|--------|------|---------|-----|
| 1  | <b>Packing Time study</b>                                                                  |      |       |      |          |        |      |         |     |
| 2  | Note: sheet records time taken by a specific person to finish filling one packet of 10 gum |      |       |      |          |        |      |         |     |
| 3  | Readings                                                                                   | Evan | Tessa | Lisa | Madaline | Morgan | Iris | Kristen | Amy |
| 5  | 1                                                                                          | 10   | 16    | 27   | 22       | 14     | 44   | 17      | 24  |
| 6  | 2                                                                                          | 17   | 20    | 25   | 16       | 13     | 24   | 24      | 32  |
| 7  | 3                                                                                          | 19   | 19    | 22   | 30       | 15     | 20   | 27      | 35  |
| 8  | 4                                                                                          | 17   | 17    | 25   | 16       | 13     | 16   | 21      | 27  |
| 9  | 5                                                                                          | 22   | 22    | 21   | 12       | 12     | 17   | 18      | 40  |
| 10 | 6                                                                                          | 17   | 13    | 27   | 34       | 15     | 16   | 26      | 28  |
| 11 | 7                                                                                          | 16   | 25    | 27   | 12       | 22     | 20   | 22      | 27  |
| 12 | 8                                                                                          | 16   | 24    | 22   | 18       | 14     | 16   | 25      | 29  |
| 13 | 9                                                                                          | 15   | 15    | 22   | 10       | 12     | 20   | 24      | 40  |
| 14 | 10                                                                                         | 14   | 18    | 23   | 20       | 14     | 15   | 20      | 36  |
| 15 | 11                                                                                         | 14   | 16    | 19   | 20       | 13     | 18   | 24      | 45  |
| 16 | 12                                                                                         | 16   | 21    | 10   | 19       | 15     | 21   | 22      | 40  |
| 17 | 13                                                                                         | 16   | 35    | 20   | 19       | 28     | 18   | 21      | 36  |
| 18 | 14                                                                                         | 18   | 43    | 22   | 12       | 13     | 19   | 20      | 41  |
| 19 | 15                                                                                         | 18   | 18    | 22   | 28       | 13     | 18   | 30      | 29  |
| 20 | 16                                                                                         | 19   | 21    | 21   | 14       | 14     | 15   | 27      | 35  |
| 21 | 17                                                                                         | 15   | 21    | 22   | 14       | 22     | 17   | 21      | 37  |
| 22 | 18                                                                                         | 16   | 17    | 23   | 24       | 12     | 19   | 21      | 43  |
| 23 | 19                                                                                         | 18   | 18    | 20   | 37       | 13     | 22   | 26      | 40  |
| 24 | 20                                                                                         | 15   | 22    | 20   | 10       | 12     | 18   | 27      | 43  |
| 25 | 21                                                                                         | 21   | 39    | 29   | 15       | 13     | 17   | 19      | 54  |
| 26 | 22                                                                                         | 20   | 19    | 27   | 7        | 14     | 18   | 22      | 26  |

|    | A                                                                            | B   | C    | D     | E    | F        | G      | H    | J   |
|----|------------------------------------------------------------------------------|-----|------|-------|------|----------|--------|------|-----|
| 1  | <b>Cleaning Time study</b>                                                   |     |      |       |      |          |        |      |     |
| 2  | Note: This sheet records time taken by a specific person to clean 10 gummies |     |      |       |      |          |        |      |     |
| 3  | Readings                                                                     | Ana | Evan | Tessa | Lisa | Madaline | Morgan | Iris | Amy |
| 5  | 1                                                                            | 14  | 52   | 34    | 38   | 32       | 15     | 23   | 75  |
| 6  | 2                                                                            | 16  | 47   | 29    | 51   | 30       | 14     | 24   | 62  |
| 7  | 3                                                                            | 18  | 51   | 50    | 23   | 28       | 21     | 31   | 31  |
| 8  | 4                                                                            | 17  | 51   | 35    | 30   | 23       | 9      | 27   | 30  |
| 9  | 5                                                                            | 14  | 58   | 21    | 35   | 44       | 13     | 30   | 69  |
| 10 | 6                                                                            | 15  | 42   | 27    | 39   | 52       | 14     | 32   | 75  |
| 11 | 7                                                                            | 27  | 67   | 21    | 47   | 19       | 12     | 26   | 62  |
| 12 | 8                                                                            | 22  | 54   | 23    | 34   | 36       | 17     | 39   | 79  |
| 13 | 9                                                                            | 26  | 42   | 39    | 23   | 40       | 10     | 24   | 95  |
| 14 | 10                                                                           | 19  | 45   | 26    | 30   | 35       | 10     | 33   | 120 |
| 15 | 11                                                                           | 16  | 58   | 28    | 40   | 47       | 10     | 34   | 69  |
| 16 | 12                                                                           | 20  | 35   | 23    | 20   | 47       | 18     | 32   | 35  |
| 17 | 13                                                                           | 14  | 44   | 25    | 27   | 19       | 14     | 36   | 78  |
| 18 | 14                                                                           | 16  | 48   | 37    | 30   | 24       | 19     | 43   | 34  |
| 19 | 15                                                                           | 17  | 52   | 25    | 34   | 41       | 18     | 32   | 73  |
| 20 | 16                                                                           | 19  | 39   | 22    | 18   | 26       | 17     | 30   | 60  |
| 21 | 17                                                                           | 18  | 57   | 20    | 30   | 41       | 15     | 59   | 79  |
| 22 | 18                                                                           | 21  | 37   | 19    | 46   | 45       | 16     | 23   | 75  |
| 23 | 19                                                                           | 19  | 44   | 28    | 29   | 30       | 9      | 31   | 68  |
| 24 | 20                                                                           | 16  | 39   | 13    | 32   | 32       | 22     | 36   | 29  |
| 25 | 21                                                                           | 24  | 42   | 13    | 27   | 15       | 20     | 29   | 62  |
| 26 | 22                                                                           | 18  | 57   | 16    | 26   | 28       | 16     | 48   | 33  |

Figure 1

Figure 2

Figure 2: Sample Time Log – Cleaning 10 Gummies

|    | A                          | B                                                                             | C           | D            | E           | F               | G             | H           | J          |
|----|----------------------------|-------------------------------------------------------------------------------|-------------|--------------|-------------|-----------------|---------------|-------------|------------|
| 1  | <b>Cleaning Time study</b> |                                                                               |             |              |             |                 |               |             |            |
| 2  | Note:                      | <b>This sheet records time taken by a specific person to clean 10 gummies</b> |             |              |             |                 |               |             |            |
| 3  |                            |                                                                               |             |              |             |                 |               |             |            |
| 4  | <b>Readings</b>            | <b>Ana</b>                                                                    | <b>Evan</b> | <b>Tessa</b> | <b>Lisa</b> | <b>Madaline</b> | <b>Morgan</b> | <b>Iris</b> | <b>Amy</b> |
| 5  | 1                          | 14                                                                            | 52          | 34           | 38          | 32              | 15            | 23          | 75         |
| 6  | 2                          | 16                                                                            | 47          | 29           | 51          | 30              | 14            | 24          | 62         |
| 7  | 3                          | 18                                                                            | 51          | 50           | 23          | 28              | 21            | 31          | 31         |
| 8  | 4                          | 17                                                                            | 51          | 35           | 30          | 23              | 9             | 27          | 30         |
| 9  | 5                          | 14                                                                            | 58          | 21           | 35          | 44              | 13            | 30          | 69         |
| 10 | 6                          | 15                                                                            | 42          | 27           | 39          | 52              | 14            | 32          | 75         |
| 11 | 7                          | 27                                                                            | 67          | 21           | 47          | 19              | 12            | 26          | 62         |
| 12 | 8                          | 22                                                                            | 54          | 23           | 34          | 36              | 17            | 39          | 79         |
| 13 | 9                          | 26                                                                            | 42          | 39           | 23          | 40              | 10            | 24          | 95         |
| 14 | 10                         | 19                                                                            | 45          | 26           | 30          | 35              | 10            | 33          | 120        |
| 15 | 11                         | 16                                                                            | 58          | 28           | 40          | 47              | 10            | 34          | 69         |
| 16 | 12                         | 20                                                                            | 35          | 23           | 20          | 47              | 18            | 32          | 35         |
| 17 | 13                         | 14                                                                            | 44          | 25           | 27          | 19              | 14            | 36          | 78         |
| 18 | 14                         | 16                                                                            | 48          | 37           | 30          | 24              | 19            | 43          | 34         |
| 19 | 15                         | 17                                                                            | 52          | 25           | 34          | 41              | 18            | 32          | 73         |
| 20 | 16                         | 19                                                                            | 39          | 22           | 18          | 26              | 17            | 30          | 60         |
| 21 | 17                         | 18                                                                            | 57          | 20           | 30          | 41              | 15            | 59          | 79         |
| 22 | 18                         | 21                                                                            | 37          | 19           | 46          | 45              | 16            | 23          | 75         |
| 23 | 19                         | 19                                                                            | 44          | 28           | 29          | 30              | 9             | 31          | 68         |
| 24 | 20                         | 16                                                                            | 39          | 13           | 32          | 32              | 22            | 36          | 29         |
| 25 | 21                         | 24                                                                            | 42          | 13           | 27          | 15              | 20            | 29          | 62         |
| 26 | 22                         | 18                                                                            | 57          | 16           | 26          | 28              | 16            | 48          | 33         |

*Figure 2: Sample Time Log – Cleaning 10 Gummies*

The data revealed that the average time to clean 10 gummies was 31.6 seconds with noticeable variation among workers, indicating inconsistency in execution and possible ergonomic or workflow issues. Packing 10 gummies took an average of 21.5 seconds and was relatively more consistent, although occasional delays occurred due to material availability and shared resources.

The collected time data served as a foundation for identifying variability, establishing performance baselines, and detecting inefficiencies within the process. By analyzing trends and standard deviations in task durations, we were able to highlight areas where standardization and layout adjustments could significantly improve overall efficiency. This analysis directly informed the root cause identification during the Analyze phase and supported targeted improvement actions in the subsequent stages of the project.

## KEY PERFORMANCE INDICATORS

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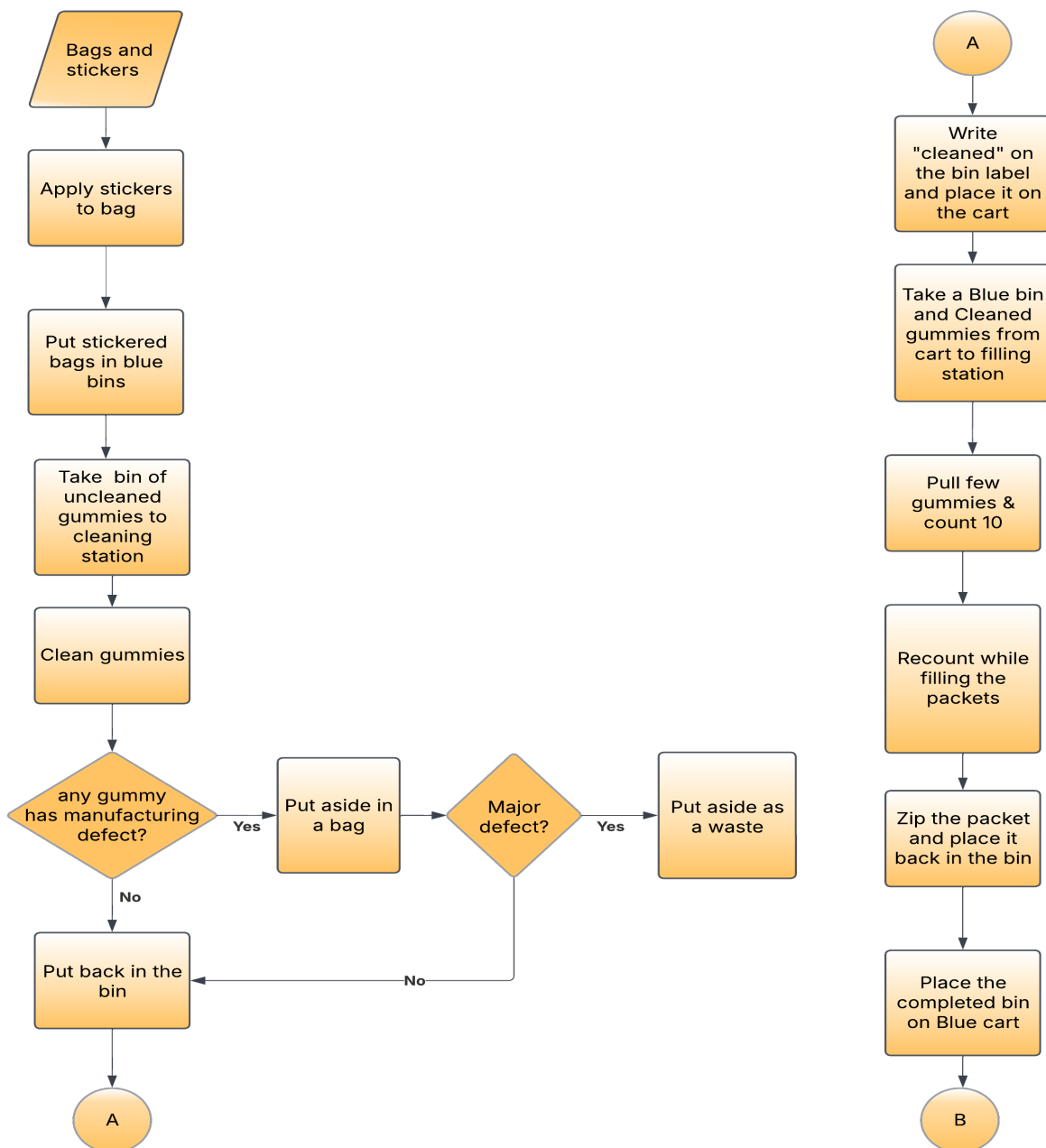
The following Key Performance Indicators (KPIs) were selected to capture critical performance areas within the packaging operation. Each KPI aligns with a core department and was used to track measurable progress throughout the DMAIC process

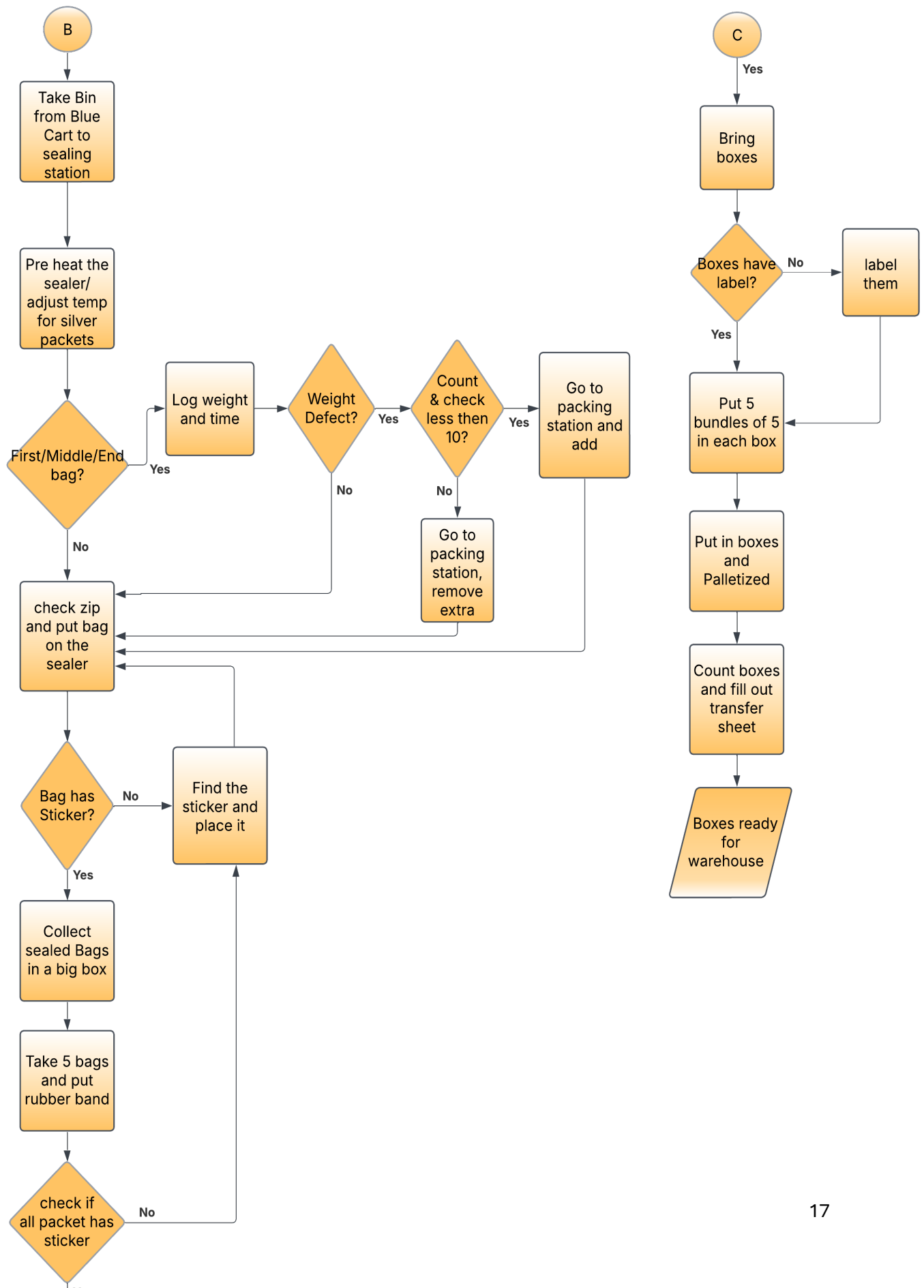
| Department | KPIs                                       |
|------------|--------------------------------------------|
| Production | Packaging Productivity (Units/hour/person) |
| Cleaning   | Cleaning Time per Bin (Minutes/bin)        |
| Packing    | Packing Time per Bin (Minutes/bin)         |
| Production | Production Cost per Packet (\$)            |
| Packing    | Worker's performance Rankings              |

This KPI tree helped ensure alignment between project goals and measurable process outputs. It served as a strategic guide throughout the DMAIC cycle by clearly connecting high-level objectives—like increasing throughput or reducing waste—to department-specific indicators. By tracking these KPIs, we were able to evaluate not only process performance but also the impact of our changes in real-time. This structure supported informed decision-making and continuous improvement

## PROCESS FLOW CHART

The following flow charts illustrate the end-to-end process followed at the Ayrloom facility for THC-infused gummy packaging. This visual representation maps each operational step from initial stickering and cleaning through filling, sealing, boxing, and palletizing.

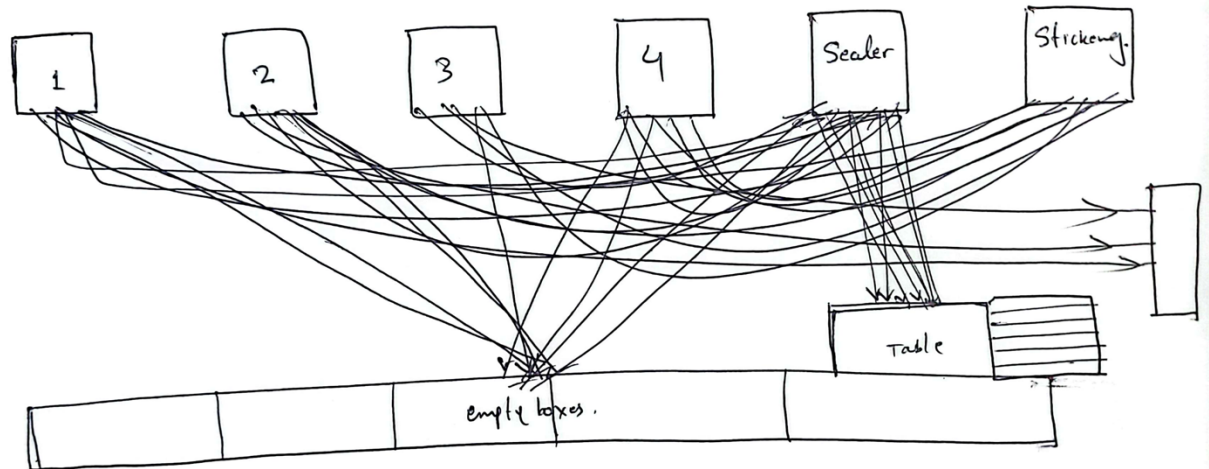




## SPAGHETTI DIAGRAM

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The spaghetti diagram below was developed to visualize the flow of materials and workers within the gummy packaging area. Each line represents a movement between stations—including packing, sealing, stickering, and resource stations like empty boxes and dirty bins.



The diagram highlights significant inefficiencies in the existing layout. Frequent, overlapping movement paths were observed, particularly around shared resources and between packaging stations and the sealing or stickering areas. This resulted in **congested walkways, frequent back-and-forth motion, and workers often bumping into each other**, leading to delays and potential safety risks.

This high-density flow pattern indicated the need for a layout redesign for material delivery, restricting worker movement, and optimizing resource placement—steps that directly contributed to smoother operations and improved productivity.

## ANALYZE PHASE

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The primary goal during this phase is to determine the root causes behind current performance gaps and process bottlenecks, backed by data analysis and observational evidence.

Ayrloom is a rapidly growing cannabis edibles brand within the Beak & Skiff umbrella. With a consistent weekly increase of 2-3% in gummy production volume, it has become critical to ensure that the packaging operations can keep pace without increasing workforce size. Our observations, time studies, and data analysis were used to better understand the current state and form the basis for improvement recommendations. This report explains our investigative approach and provides detailed findings that will help drive sustainable improvements.

### OBJECTIVE OF THE ANALYZE PHASE

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The Analyze phase serves as a bridge between identifying issues (Define and Measure phases) and generating solutions (Improve phase). The primary purpose of this phase is to investigate why the problems identified in the Measure phase are occurring. Specifically, our objectives were:

- To identify the key variables that cause delays and inefficiencies in the gummy packaging process.
- To quantify the impact of variability in worker performance.
- To isolate process steps that create bottlenecks or excessive wait times.
- To apply Six Sigma tools such as the Fishbone Diagram, 5 Whys, Pareto Charts, and Time Studies to visualize and understand cause-effect relationships.

By achieving these objectives, we aimed to inform actionable strategies for improving throughput, reducing rework, and enhancing operational consistency.

## TIME STUDY SUMMARY

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One of the core analytical activities in the Analyze phase was conducting time studies for major tasks performed in the packaging line. Our team collected detailed time data for three key tasks:

- **Packing** (placing 10 gummies in a pouch)
- **Cleaning** (removing visible defects or sticky residues from gummies)
- **Stickering** (applying regulatory labels to packaging pouches)

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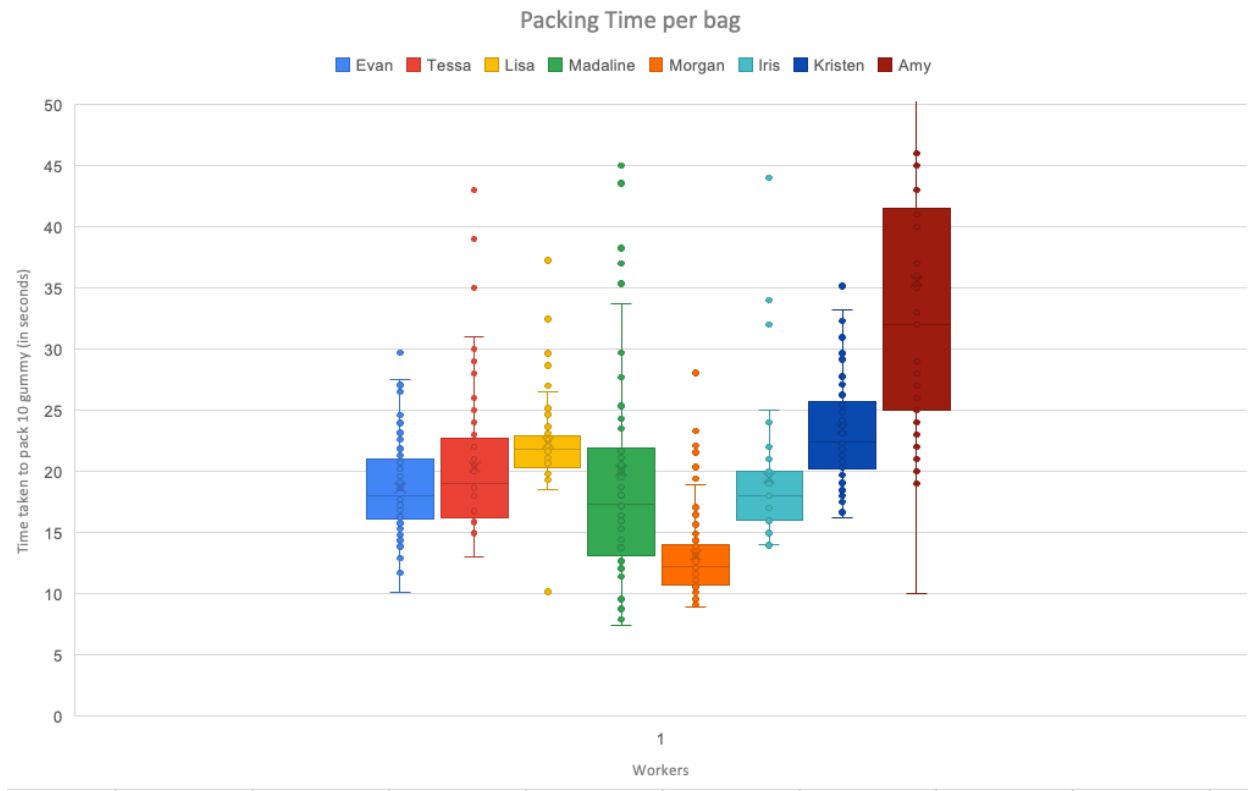
### PACKING TIME OBSERVATIONS

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The packing process is performed entirely manually. Workers are responsible for counting out exactly 10 gummies by hand, placing them into a packaging pouch, and then closing the pouch's zip-lock seal. After sealing, each filled pouch is placed into a blue bin for subsequent processing. This labor-intensive task requires both precision and speed, making performance consistency vital.

Using the "Packing Data" sheet, we recorded multiple rounds of timing per worker. Results showed substantial performance disparities across employees. Morgan emerged as the fastest worker, consistently packing gummies with a median time of just over 10 seconds per bag. In contrast, Amy exhibited the slowest performance, often taking 35 to 50 seconds for the same task.

This variability is critical because it demonstrates that certain employees could potentially produce 80 bags per hour, while others struggle to meet half of that target. Such disparities highlight a need for strategic task allocation based on performance.



### WORKER ANALYSIS (PACKING TIME PER BAG)

- **Fastest & Most Consistent Packers:**
  - **Morgan** (orange) had the **lowest median packing time** (~12 seconds) and a very **tight spread**, making her the most efficient and reliable packer.
  - **Madaline** (green) also performed well with a low median and minimal variation, close behind Morgan.
- **Moderate Performers:**
  - **Evan, Tessa, Lisa, and Iris** all had **median times in the 18–23 second range**, with relatively moderate spread, suggesting steady performance.
- **Slower Packers:**
  - **Kristen and Amy** had the **highest median times**, with Amy's around 33 seconds and Kristen's slightly below that. Amy also showed a wide IQR, indicating inconsistency.

- **Outliers:**

- Several workers (e.g., Amy, Madaline, Lisa) had outliers, but in Amy's case, it was more frequent and toward the slower end, potentially affecting overall productivity.

*The image below shows the final output of this packing process: a transparent bin filled with neatly stickered, pouches of Ayrloom gummies. These bins are stacked on top of one another and placed on a worktable, ready to be moved to the sealing or boxing stage. This snapshot visually captures the result of manual labor and highlights the need for organized and ergonomic handling as the volume continues to grow.*



#### CLEANING TIME OBSERVATIONS

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Cleaning involves retrieving a full bin of gummies directly from cold storage. These gummies, especially sour varieties, are visibly sticky and frequently coated in excess sugar or debris, as seen in the image of unclean gummies stored on metal trays. These trays are housed in storage racks and hold multiple layers of freshly made gummies awaiting the cleaning stage.



*The debris, clumps, and sugar accumulation make manual cleaning both necessary and time-consuming, highlighting the challenge of maintaining quality while meeting speed expectations.*

Each gummy must be manually inspected and wiped clean. Workers use their hands to remove surface moisture or sugary clumps. If a gummy is visibly defective (e.g., misshaped, torn, fused, or overly sticky), it is removed and placed into a "defective" bin for rework or disposal. This inspection and wiping process varies widely among employees.

The cleaning process is currently performed without any formalized tools or SOPs, though some use of plastic scoops or trays was observed. Ana delivered the best performance, averaging under 20 seconds per 10-gummy batch. Conversely, Amy's

cleaning time frequently exceeded 80 seconds per batch. This more than 4x time discrepancy between workers introduces a significant throughput constraint.

Such extreme variation reduces predictability in shift scheduling and adds pressure on downstream processes like packing and sealing. From our time studies and visual observations, it is evident that cleaning is not only the most inconsistent task but also the largest contributor to packaging delays and system-wide bottlenecks.

Cleaning involves retrieving a full bin of gummies directly from cold storage. These gummies, often sticky—especially in the case of sour varieties—must be manually inspected and wiped clean. Each worker uses their hands to remove surface moisture or residue. If a gummy is visibly defective (e.g., misshaped, torn, or fused to another), it is separated and placed into a "defective" bin for rework or disposal.

This process is performed manually and without a formalized tool, though some use of plastic scoops and trays was observed. Despite the seeming simplicity of this task, it proved to be the most time-consuming and inconsistent. Ana delivered the best performance, averaging under 20 seconds per 10-gummy cleaning batch. Amy, however, often exceeded 80 seconds per batch.

This extreme variation not only drags down average throughput but also introduces uncertainty in scheduling and planning. It was evident from our data that the cleaning step is the largest contributor to bottlenecks in the system.

## STICKERING PERFORMANCE

Once packaging pouches are ready, they must be labeled with compliance and product information stickers. This is done manually. Workers receive sticker sheets—each sheet containing 154 pre-printed adhesive stickers. Using their hands, they peel off each sticker and carefully apply it to a flat area on the pouch, specifically to the right of the barcode.

The image below illustrates an example of un-stickered packaging pouches, stacked neatly and ready to be labeled. The designated space for sticker placement—next to the barcode—is critical for regulatory compliance and product traceability. Since stickers include essential information like QR codes, batch IDs, THC content, and warning labels, accurate and consistent placement is essential.



In the "Stickering" data, we analyzed Anna's performance across three recorded sessions. She applied 154 stickers in just under 10 minutes, averaging approximately 3.5 seconds per sticker. This task, while repetitive, showed minimal variation and posed no significant constraint in the overall process.

Nonetheless, stickering represents an ideal task for support role delegation. It is best assigned to workers with slower packing or cleaning times so that higher-performing individuals can remain focused on bottlenecked tasks. The clarity of sticker placement also contributes to product presentation and customer trust, reinforcing the importance of training and visual quality checks even for this seemingly minor step.

Once packaging pouches are ready, they must be labeled with compliance and product information stickers. This is done manually. Workers receive sticker sheets—each sheet containing 154 pre-printed adhesive stickers. Using their hands, they peel off each sticker and carefully apply it to a flat area on the pouch.

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## ROOT CAUSE ANALYSIS

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Root cause analysis is essential in the Analyze phase to diagnose the true underlying reasons for the problems observed in the process. While symptoms such as "slow packaging" or "low throughput" are visible on the surface, Lean Six Sigma emphasizes drilling deeper to reveal what causes these issues. To conduct this investigation, we used two widely recognized tools: the **Fishbone Diagram (Ishikawa)** and the **5 Whys Analysis**. These tools not only helped visualize the relationship between potential causes and the problem but also guided our team in distinguishing between root causes and superficial ones.

## FISHBONE (ISHIKAWA) DIAGRAM

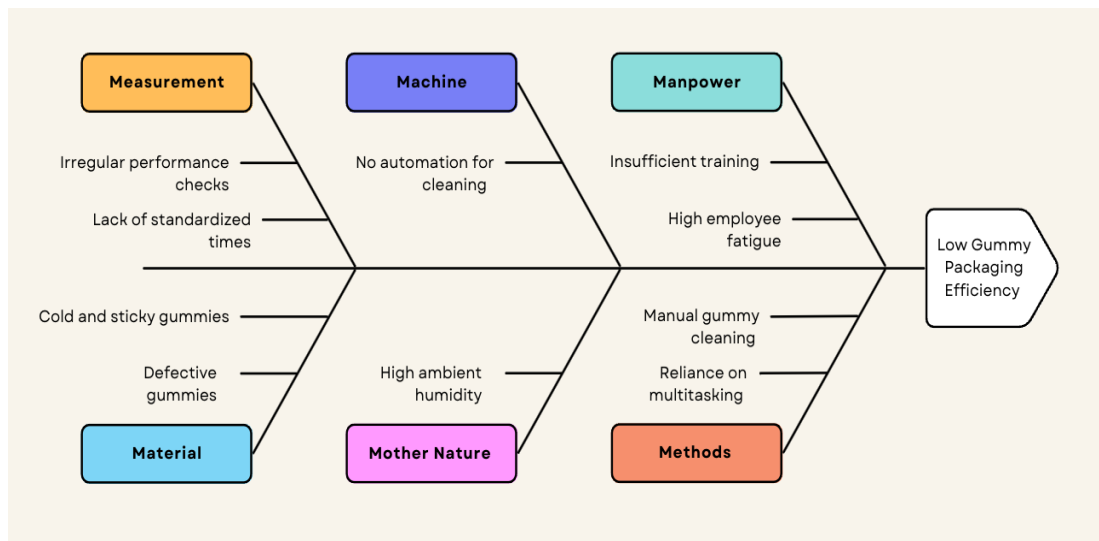
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A Fishbone Diagram, also known as a cause-and-effect diagram, is a visual tool that categorizes potential causes of problems into logical groupings. In our case, the main problem or effect was defined as "Packing Productivity Below Target."

We identified and grouped contributing factors into six key categories:

- **Manpower:** Workers operate at drastically different speeds; some are extremely fast and consistent, while others are slow and erratic. Training is informal, and many new employees learn by shadowing rather than with standardized procedures.
- **Methods:** There are no standard operating procedures (SOPs) for tasks like cleaning, stickering, or handling sour gummies. As a result, workers develop their own habits, leading to high process variation. One significant but currently out-of-scope contributor is the way gummy slurry is poured into molds in the kitchen. Instead of using automated nozzles to fill each mold cavity directly, the slurry is poured manually in bulk and spread by scrapper. This uneven distribution leads to clumped, misshapen, or sticky gummies that require more intensive manual cleaning downstream. While this upstream step was not part of our formal project scope, its impact on the cleaning workload was noted during our observations.
- **Machines:** Manual sealing adds time and is inconsistent. Manual processes dominate several tasks in the packaging workflow. For example, the use of basic tools or bare hands for cleaning, combined with manual handling of trays and packaging materials, increases fatigue and variation. There is a lack of ergonomic equipment to support repetitive actions, such as cleaning or transferring gummies.
- **Materials:** Sour gummies are sticky and take longer to clean. Additionally, packaging materials such as stickers are handled manually and fetched individually rather than being organized in advance.
- **Measurements:** Time tracking is limited. Workers are not consistently monitored, and no benchmarks or KPIs are posted to set expectations or encourage goal setting.

- **Environment (Mother Nature):** The packaging room lacks an efficient layout. Workers must walk to collect gloves, bins, stickers, and cleaned items. This unnecessary motion contributes to lost time and confusion.



This diagram provided a structured way to brainstorm all potential root causes and enabled the team to identify recurring patterns across tasks and roles.

## 5 WHYS ANALYSIS

To further understand the inconsistency in cleaning performance, we applied the 5 Whys analysis. This method is used to drill down from the symptoms of a problem to its root cause by asking "Why?" iteratively.

**Problem Statement:** Cleaning times vary greatly across workers, leading to inconsistent throughput and delays in downstream processes.

- **Why 1:** *Because some workers take significantly longer to clean gummies than others.*

- **Why 2:** *Because each worker uses a different method for cleaning, with no standardization in the tools or techniques used.*
- **Why 3:** *Because there are no documented standard operating procedures (SOPs) or visual job aids that define a preferred method.*
- **Why 4:** *Because cleaning was treated as an extra task rather than a core activity during packaging process design.*
- **Why 5:** *Because the initial focus of the packaging operation was on maximizing output to meet growing product demand, and less attention was given to designing structured, replicable processes for non-automated steps.*

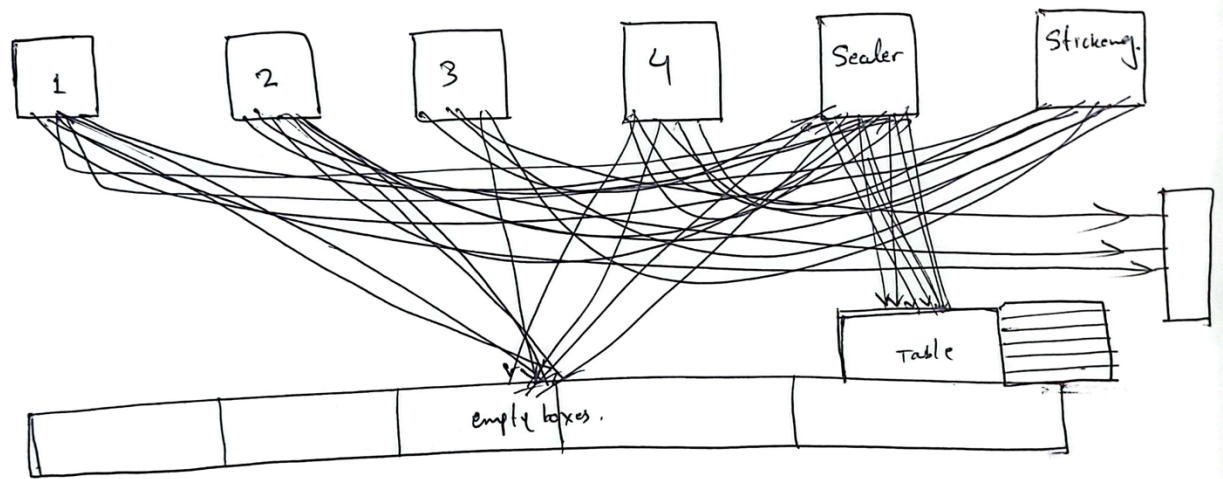
This refined analysis shows that inconsistency in cleaning performance stems not just from individual skill, but from systemic gaps in training and process definition.

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#### LAYOUT AND FLOW ANALYSIS (SPAGHETTI DIAGRAM)

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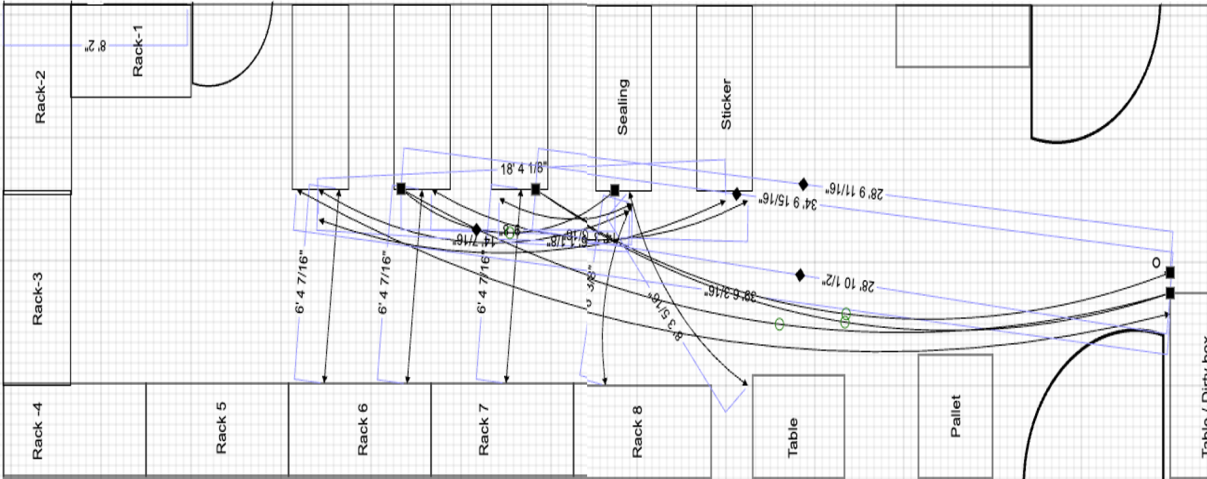
Efficient layout and workflow are foundational principles in Lean manufacturing. At Ayrloom, we identified that poor spatial organization and excessive motion were significantly contributing to the inefficiencies within the packaging process. Our direct observations, interviews, and spaghetti diagram analysis revealed that workers frequently left their stations during operations whether to collect gloves, retrieve clean bins, find sticker sheets, or carry sealed pouches to the next area. These unnecessary motions introduce wasted time, disrupt workflow, and cause bottlenecks.



The spaghetti diagram of the current packaging area highlights an erratic, non-linear flow of people and materials. The movement paths of employees intersect repeatedly across various stations (from Station 1 to Sticker and Sealing), indicating frequent backtracking and role switching. Several flow lines overlap in the center of the room, demonstrating congestion and redundant paths between racks, tables, and work areas. Shared access to dirty box tables and supply racks contributes to collisions in workflow.

Specific observations include:

- Workers routinely walk 6–18 feet between task zones (based on distance markings on the layout).
- All stations access the same bins, tables, and racks, causing overlap and delays.
- The current flow does not follow a logical path such as U-shaped or straight-line assembly, increasing the cognitive load and variability.
- No clear spatial division exists between cleaning, packing, and sealing zones.



To address these issues, we recommend the following redesign elements:

- **Dedicated Stations by Function:** Implement strict separation of cleaning, packing, sealing, and boxing stations, with each equipped with its own supply stock.
- **Central Supply and Dirty bin Zones:** Relocate supplies and waste collection points to the periphery, ensuring minimal interference with the primary workflow.
- **Introduce a Runner Role:** This individual ensures material replenishment and waste clearing, keeping packers and cleaners focused on task execution.
- **Establish a Linear or U-Shaped Layout:** Redesign flow from left to right (or U-shaped) so that workers and materials progress in a single direction with no backtracking.
- **Use Visual Controls:** Apply colored floor tape and workstation signage to clearly indicate roles, routes, and boundaries for movement.

From our evaluation, these changes are expected to reduce motion waste by 15–25% and improve task clarity, especially for new employees. Implementing a Lean-aligned layout is critical for scaling the packaging process while maintaining consistency and speed.

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observations and interviews revealed that workers frequently left their stations during operations — whether to collect gloves, retrieve clean bins, find sticker sheets, or carry sealed pouches to the next area. These unnecessary motions introduce wasted time, disrupt workflow, and cause bottlenecks.

For example, during a standard 10-minute observation of the packaging floor, one worker spent over 3 minutes simply walking between tasks or waiting for materials to arrive. When this behavior is scaled across multiple shifts and workers, the cumulative motion waste becomes a significant drag on productivity. Moreover, many resources (like sticker sheets, gloves, and boxes) are not centrally located, causing interruptions even in high-performing workers' routines.

To address this, we conducted a detailed process walk and mapped the physical flow of materials and personnel across the current layout. This was compared with ideal Lean principles, which emphasize minimizing motion, creating clear stations for each task, and enabling smooth transitions between sequential activities.

As a result, we propose the following redesign elements:

- **Designated Cleaning, Packing, and Sealing Zones:** Each zone should have its own supplies and dedicated bins, preventing cross-traffic and confusion.
- **Centralized Material Cart with a Dedicated Runner:** A runner will be responsible for stocking sticker sheets, gloves, clean bins, and retrieving filled boxes. This ensures that core packaging staff remain focused on their primary task without leaving their stations.
- **One-Way Process Flow:** Layout should follow a U-shaped or straight-line design from cleaning → packing → sealing → boxing. This eliminates backtracking and prevents workers from crossing paths.
- **Visual Cues and Labeling:** Floor tape, signs, and standardized workstations should be implemented to define zones, tool locations, and finished goods areas.

By improving the physical environment, Ayrloom can reduce motion waste by an estimated 15–20%, based on industry benchmarks. Combined with proper task allocation, these changes will lead to a smoother, faster, and more scalable packaging operation.

We observed significant motion waste due to workers leaving their stations frequently—to collect stickers, bring gloves, fetch new bins, or move sealed pouches. The current packaging room lacks a standardized layout, creating unnecessary travel and slowing down work.

A revised layout plan that incorporates a centralized runner, designated sticker station, and linear flow between tasks will reduce non-value-added motion and streamline throughput.

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### WORKER PERFORMANCE RANKING

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The worker time study conducted as part of the Analyze Phase revealed significant differences in task execution speeds across employees. To ensure process efficiency and maximize throughput, aligning tasks to each worker’s strengths is essential.

Below is a detailed performance table compiled from observational data for packing, cleaning, and stickering tasks:

| <b>Worker</b>  | <b>Avg. Packing Time (secs)</b> | <b>Avg. Cleaning Time (secs)</b> | <b>Stickering Time (sec/sticker)</b> | <b>Suggested Roles</b> |
|----------------|---------------------------------|----------------------------------|--------------------------------------|------------------------|
| <i>Morgan</i>  | 10.5                            | 14.9                             | -                                    | Core Packing           |
| <i>Ana</i>     | -                               | 19.5                             | -                                    | Core Cleaning          |
| <i>Iris</i>    | 14.0                            | 24.3                             | -                                    | Support Packing        |
| <i>Tessa</i>   | 20.5                            | 32.1                             | -                                    | Support Packing        |
| <i>Evan</i>    | 22.1                            | 35.0                             | -                                    | Support Cleaning       |
| <i>Kristen</i> | 30.2                            | 49.8                             | -                                    | Stickering/Support     |
| <i>Amy</i>     | 45.6                            | 81.2                             | 3.5                                  | Stickering/Support     |

## Key Observations:

- **Morgan** is the most efficient packer with an average of 10.5 seconds per bag and should be dedicated to high-speed packing roles.
- **Ana** consistently delivered the best cleaning times (19.5 seconds per batch), making her a great fit for dedicated cleaning tasks.
- **Amy** had the highest variability and longest times in both packing and cleaning but demonstrated consistency in stickering (~3.5 seconds per label), suggesting her ideal role is in supportive tasks like labeling and boxing.
- **Kristen**, while slower in manual work, may be more effective in standardized, repetitive work like labeling.

By leveraging this data, we propose a performance-based work allocation strategy where:

- Top performers focus on value-generating core tasks.
- Moderate performers assist with preparation, rework, and boxing.
- Lower-performing individuals are placed in roles that require consistency over speed (e.g., stickering).
- This division of labor will reduce cycle time variation, minimize errors, and create a more balanced workload across the team, aligning well with Lean principles of waste reduction and role clarity.

Based on average cycle time across tasks:

- **Top Performers:** Morgan, Ana, Iris
- **Moderate Performers:** Tessa, Evan
- **Low Performers:** Amy, Kristen

Assigning roles based on this ranking allows for specialization. High performers can focus on core tasks like packing and cleaning, while others support with stickering or boxing.

## SUMMARY OF FINDINGS

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The findings from the Analyze Phase provide critical insight into the underlying inefficiencies within Ayrloom's packaging workflow. By triangulating quantitative time study data, qualitative floor observations, and root cause tools like Fishbone and Pareto, we observed the following key issues:

- **Cleaning as the Primary Bottleneck:** Cleaning operations have the highest cycle time and variability across all packaging activities. The slowest worker takes up to 80 seconds per 10-gummy batch, compared to the fastest at under 20 seconds. This gap leads to queueing delays and underutilization of packers.
- **Worker Speed Variation:** There is a 4x difference in task execution speed between top and bottom performers. Inconsistent performance leads to erratic daily output and makes it difficult to meet scaling demand.
- **Manual and Redundant Processes:** All major tasks — cleaning, packing, sealing, and stickering — are performed manually. Motion studies show significant time wasted on non-value-added steps, such as retrieving supplies or backtracking across the room.
- **Lack of Visual Standards and SOPs:** There are no documented standard operating procedures or visual work instructions posted in workstations. New workers learn by watching others, which increases variability and risk of error.
- **Non-Optimized Floor Layout:** The current layout lacks flow logic. Movement overlaps, congestion, and supply station ambiguity increase motion waste and slow down transitions between tasks.
- **Underutilized Equipment:** Although an automatic sealing machine exists, it is rarely used due to lack of training. Manual sealing increases variability and operator fatigue.

These findings serve as the bedrock for developing targeted improvement solutions that align with Lean Six Sigma principles of waste reduction, role clarity, and standardized processes.

- Cleaning is the dominant bottleneck
- Worker speed varies 2–4x across roles
- Manual processes and movement waste slow the system
- Visual SOPs and training are lacking
- No standardized packaging layout or material flow exists

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## ANALYZE PHASE INSIGHTS

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The Analyze Phase confirmed that while Ayrloom’s workforce is dedicated and capable, systemic inefficiencies in layout, training, and task distribution limit overall throughput. Our investigation revealed that cleaning is the dominant constraint, exacerbated by variability in worker speeds and a lack of standardization in daily operations. Furthermore, excessive movement between stations and underuse of available automation introduce avoidable delays.

By using Lean Six Sigma tools such as the Fishbone Diagram, Pareto Analysis, and Spaghetti Mapping, we were able to visualize and validate the specific areas causing performance drops. Data from detailed time studies further reinforced the need for a role-based assignment strategy and better layout configuration.

The recommendations derived from this analysis are not just theoretical — they are practical, measurable, and achievable. Implementing these changes will yield:

- Increased consistency in output
- Better use of skilled labor
- Reduction in downtime and walking waste
- Enhanced onboarding process through SOPs

- Greater alignment between packaging capacity and rising production volumes

With these insights in place, the Improve Phase can now be targeted with confidence. This phase will focus on testing, validating, and scaling the proposed interventions — ensuring that the changes we recommend translate into real-world performance gains for Ayrloom’s packaging team.

The Analyze Phase validated our hypothesis that the biggest constraints at Ayrloom lie in process variation and inefficient task allocation. By using data, observational insights, and structured problem-solving tools, we were able to uncover the drivers of delay and inconsistencies in throughput. The findings provide a clear foundation to develop and implement impactful solutions in the Improve phase.

# IMPROVE PHASE

## FAILURE MODE AND EFFECTS ANALYSIS

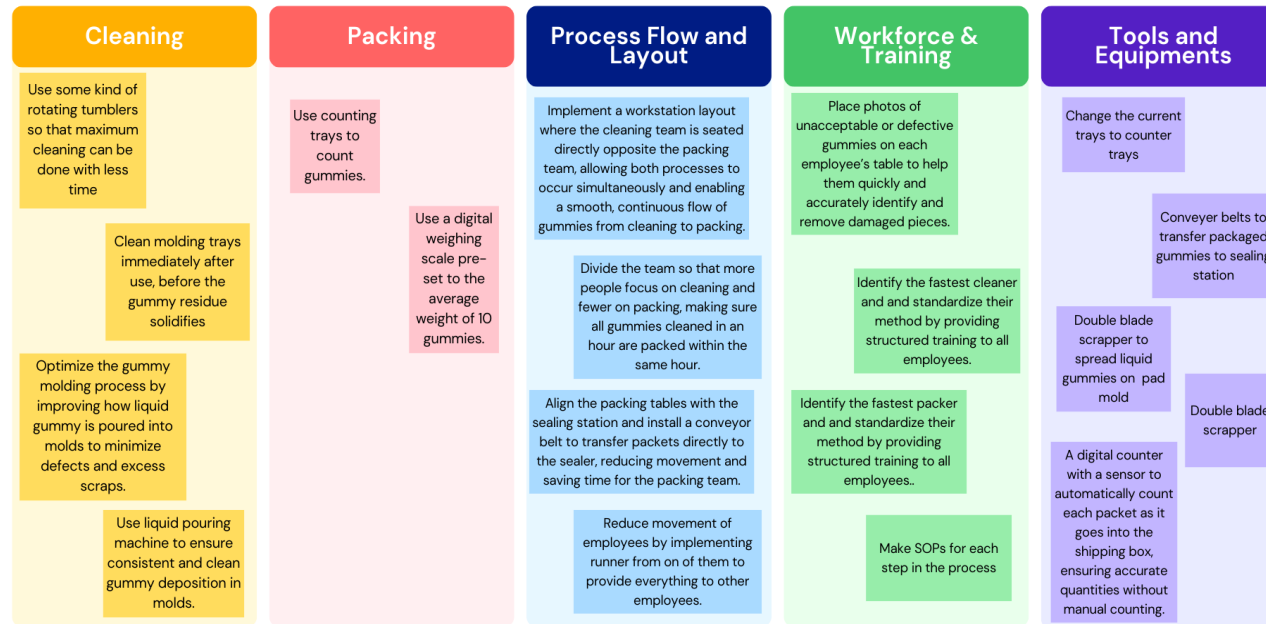
Process Name: Gummies packaging process.

| Process Step                        | Failure Mode                                         | Severity (1-10): 10=Most severe | Occurrence (1-10): 10=highest prob of occurrence | Detection (1-10): 10 = lowest probability of detection | Risk Priority Number | Improvement Action                            |
|-------------------------------------|------------------------------------------------------|---------------------------------|--------------------------------------------------|--------------------------------------------------------|----------------------|-----------------------------------------------|
| Gummy Cleaning                      | Broken or attached residual                          | 8                               | 8                                                | 6                                                      | 384                  | Train staff to identify defects better        |
| Packing 10 gummies in one packet    | Packet has less or more than 10 gummies              | 8                               | 5                                                | 6                                                      | 240                  | Introduce counting trays or semi-auto counter |
| Sealing & Quality Check             | Packet not sealed properly, or miscount not detected | 9                               | 3                                                | 6                                                      | 162                  | Increase sample check rate                    |
| Boxing 25 packets into shipping box | Incorrect total number of packets (not 25)           | 10                              | 1                                                | 2                                                      | 20                   | Digital counter / Weighing Machine            |

To systematically identify weaknesses in the current gummy packaging process, we conducted a Failure Mode and Effects Analysis (FMEA). This method allowed us to evaluate each process step by assigning scores for severity, occurrence, and detection likelihood. The resulting Risk Priority Numbers (RPNs) were used to rank failure modes based on their overall risk impact.

Our analysis revealed that the highest RPNs were associated with two critical areas: gummy cleaning and the accurate packing of 10 gummies per packet. These steps pose the greatest risk to efficiency and quality if not addressed. As a result, we are prioritizing improvements in these areas to significantly enhance process reliability and throughput.

## AFFINITY DIAGRAM



To organize and evaluate the wide range of ideas generated during the brainstorming phase by our team, we used an Affinity Diagram. This tool helped us group related improvement suggestions under common themes such as Cleaning, Packing, Process Flow and Layout, Workforce & Training, and Tools and Equipment. By clustering ideas in this way, we were able to clearly identify focus areas, uncover relationships between suggestions, and streamline the problem-solving approach. The diagram highlights targeted strategies like improving molding cleanliness, using counting tools, aligning workstations, standardizing training, and implementing automation—all aimed at increasing the speed and accuracy of the gummy packaging process.

## RECOMMENDED SOLUTIONS

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- **Solution1:** Use the liquid pouring machine to ensure consistent and clean gummy deposition in molds.
- **Solution2:** Use a digital weighing scale pre-set to the average weight of 10 gummies.
- **Solution3:** Implement a workstation layout where the cleaning team is seated directly opposite the packing team, allowing both processes to occur simultaneously and enabling a smooth, continuous flow of gummies from cleaning to packing.
- **Solution4:** Divide the team so that more people focus on cleaning and fewer on packing, making sure all gummies cleaned in an hour are packed within the same hour.
- **Solution5:** Align the packing tables with the sealing station and install a conveyor belt to transfer packets directly to the sealer, reducing movement and saving time for the packing team.
- **Solution6:** Reduce movement of employees by implementing runner from one of them to provide everything to other employees.
- **Solution7:** Place photos of unacceptable or defective gummies on each employee's table to help them quickly and accurately identify and remove damaged pieces.
- **Solution8:** Identify the fastest cleaner and standardize their method by providing structured training to all employees.
- **Solution9:** Identify the fastest packer and standardize their method by providing structured training to all employees.
- **Solution10:** Conveyor belts to transfer packaged gummies to sealing station

\*\*All the solutions mentioned above are **not ranked**; they are random solutions filtered out based on our domain knowledge and that align with our overall goal.

We cannot use all 10 solutions mentioned above due to time and resource constraints, so we tried to extract the top two to three solutions that can be implemented quickly and easily. To do that, we will use the assessment criteria in the next step.

## ASSESSMENT CRITERIA

|                                           | Ashish | Katherine | Meet | Meghanath | Sai | Weighted Total |
|-------------------------------------------|--------|-----------|------|-----------|-----|----------------|
| Can be implemented quickly and easily     | 42     | 39        | 40   | 44        | 38  | 203            |
| Will it achieve the objective             | 29     | 32        | 29   | 28        | 30  | 148            |
| Criteria Low cost or no cost to implement | 19     | 19        | 20   | 19        | 22  | 99             |
| Won't impact the customer                 | 10     | 10        | 11   | 9         | 10  | 50             |
|                                           | 100    | 100       | 100  | 100       | 100 | 500            |

To fairly evaluate and prioritize our proposed solutions, our team developed an **assessment criteria chart**. Each team member was given **100 points** to distribute across four different criteria based on how important they felt each one was. These criteria were:

1. Can it be implemented quickly and easily?
2. Will it achieve the objective?
3. Is it low-cost or free to implement?
4. Will it have no negative impact on the customer?

By collecting everyone's individual point distributions, we calculated the **weight for each criterion**. These weights reflect the team's collective priorities and were later used in our scoring system to help us decide which solutions would be the most effective and practical to implement.

## SOLUTION SELECTION MATRIX

|             | Can be implemented quickly and easily | Will it achieve the objective | Low cost or no cost to implement | Won't impact the customer | Final weighted score | Ranking |
|-------------|---------------------------------------|-------------------------------|----------------------------------|---------------------------|----------------------|---------|
| Weighting   | 0.41                                  | 0.30                          | 0.20                             | 0.10                      |                      |         |
| Solution 1  | 9                                     | 9                             | 1                                | 8                         | 7.32                 | 5       |
| Solution 2  | 4                                     | 4                             | 8                                | 8                         | 5.19                 | 10      |
| Solution 3  | 7                                     | 8                             | 9                                | 9                         | 7.89                 | 3       |
| Solution 4  | 9                                     | 8                             | 8                                | 9                         | 8.51                 | 2       |
| Solution 5  | 7                                     | 9                             | 5                                | 8                         | 7.30                 | 6       |
| Solution 6  | 10                                    | 9                             | 9                                | 9                         | 9.41                 | 1       |
| Solution 7  | 7                                     | 5                             | 6                                | 9                         | 6.41                 | 8       |
| Solution 8  | 6                                     | 8                             | 6                                | 7                         | 6.69                 | 7       |
| Solution 9  | 6                                     | 8                             | 9                                | 9                         | 7.49                 | 4       |
| Solution 10 | 5                                     | 7                             | 5                                | 8                         | 5.89                 | 9       |

To determine the most effective solutions from our list of alternatives, we used a **“Solution selection matrix”**. Each row in the matrix represents one of the **Top 10 solutions** we identified earlier. For each solution, the team assigned scores (typically from 1 to 10) based on how well it satisfies each evaluation criterion. These individual scores were then multiplied by the corresponding **weights** assigned to each criterion (from our assessment criteria exercise). The result is a **Final Weighted Score** for each solution.

To enhance clarity, we applied **color coding** to these scores—green indicates a strong (high-scoring) solution, while red indicates a weaker one. Finally, we **ranked** the solutions from highest to lowest based on their weighted scores, helping us identify which ideas to prioritize for implementation.

Based on this analysis, we identified the **top two solutions**:

1. Reduce employee movement by implementing a runner to provide everything to other employees.
2. Divide the team so that more people focus on cleaning and fewer on packing, ensuring that all gummies cleaned within an hour are packed in the same hour.

## PILOT TEST

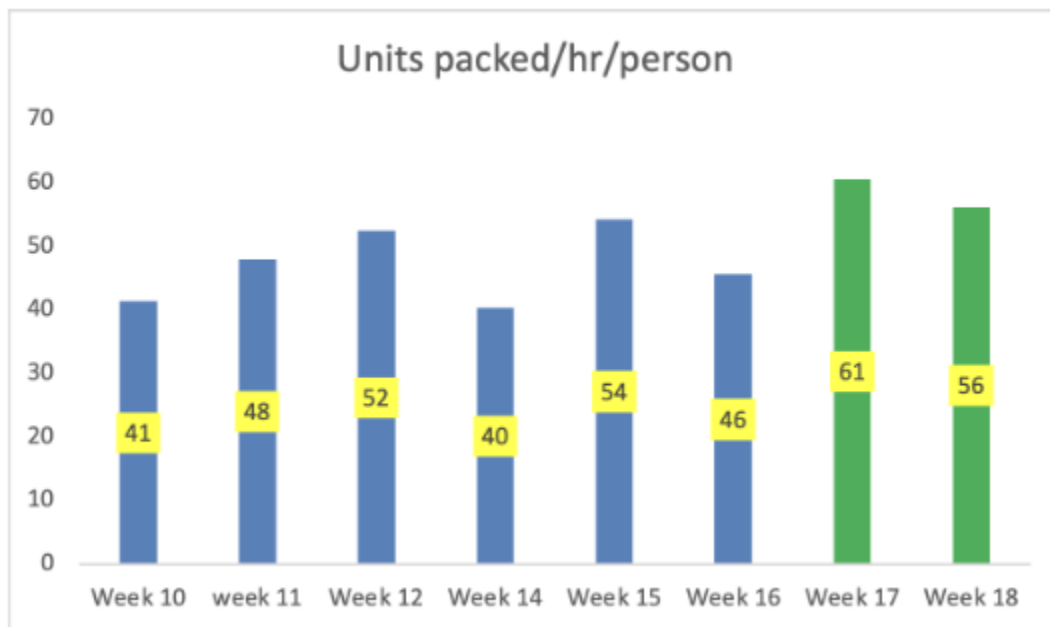
From **April 21st to May 2nd**, we conducted a pilot test for the two primary solutions mentioned above. We monitored weekly packaging productivity, measuring units packed per hour per person. The summary of the collected data is as follows:

| Week    | Units | Z-score    |
|---------|-------|------------|
| Week 10 | 13648 | -1.2553979 |
| week 11 | 15974 | -0.1664119 |
| Week 12 | 17667 | 0.62621637 |
| Week 13 | 20482 | 1.94414226 |
| Week 14 | 14218 | -0.9885354 |
| Week 15 | 16946 | 0.2886588  |
| Week 16 | 14186 | -1.0035172 |
| Week 17 | 17004 | 0.31581322 |
| Week 18 | 16840 | 0.23903175 |

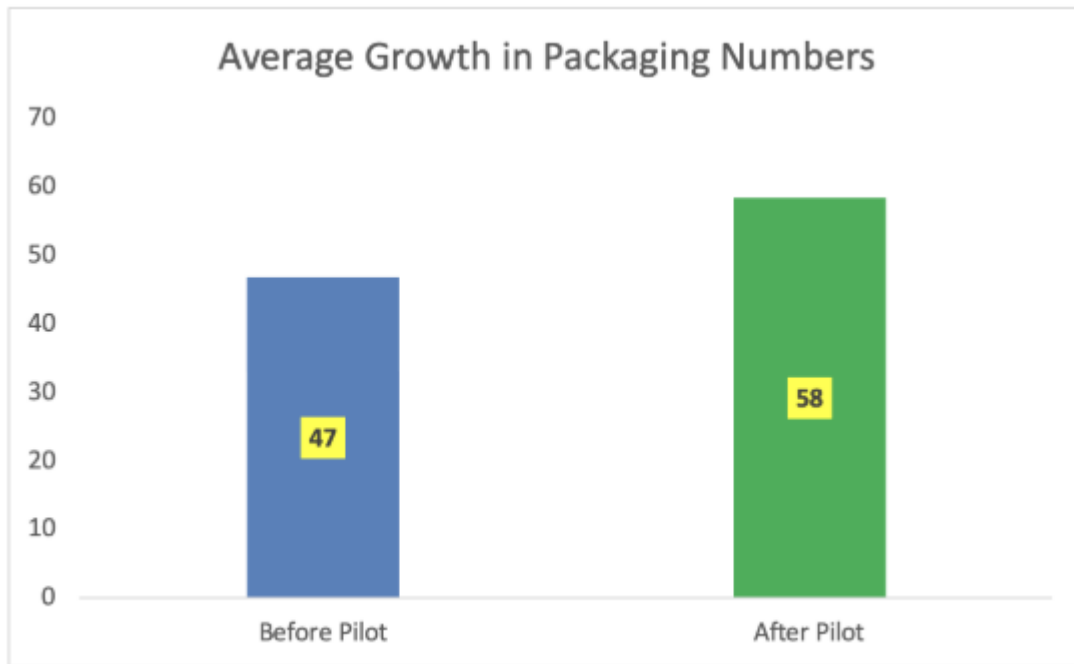


| Week    | Units packed/hr/person |
|---------|------------------------|
| Week 10 | 41.4                   |
| week 11 | 48                     |
| Week 12 | 52.4                   |
| Week 14 | 40.3                   |
| Week 15 | 54.3                   |
| Week 16 | 45.6                   |
| Week 17 | 60.6                   |
| Week 18 | 56.2                   |

In the above data, we calculated the z-score to eliminate any unusual or outlier data from our analysis.



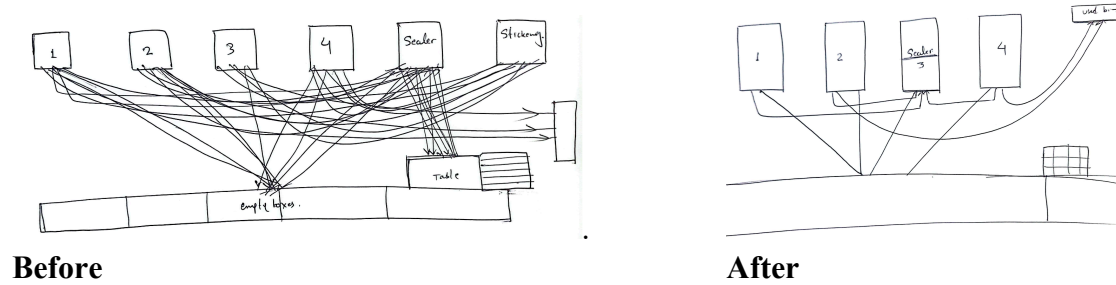
The above chart depicts the number of units packed per hour per person, with **blue** representing the productivity before the pilot test and **green** representing during the pilot test.



The chart above depicts the average number of units packed per hour per person, **with blue representing the productivity before the pilot test and green representing the period during the pilot test.**

Due to the limited number of data points after the pilot test, **a statistical test like the two-sample t-test was not performed at this stage.** Instead, we used average productivity and percentage improvement as preliminary indicators. Once more post-pilot data is available, we plan to conduct a statistical comparison to confirm long-term effectiveness.

## IMPACT OF LAYOUT REDESIGN: SPAGHETTI DIAGRAM COMPARISON



To assess and visualize the physical movement of workers during the packaging process, we used **spaghetti diagrams**. The first diagram represents the layout **before** implementing our new process, while the second diagram shows the layout **during the pilot test**.

The “before” diagram highlights excessive and inefficient movement between cleaning, packing, sealing, and boxing stations. In contrast, the “pilot” layout that includes a **“Runner”** shows a much more streamlined flow, with significantly reduced path overlaps and backtracking. This improvement was achieved by assigning runners to do all the movements instead of having other people let all employees carry out their work without wasting time on other tasks.

This visual comparison clearly supports our time and efficiency data, confirming that the layout changes **successfully minimized movement**, which directly contributed to increased productivity and reduced fatigue among workers.

## FINANCIAL ANALYSIS

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### Total Productivity per hour:

| Before                     | After                      |
|----------------------------|----------------------------|
| 1 Person = 47 packets/hour | 1 Person = 58 packets/hour |

### Cost per bag:

| Before                                                                          | After                                                                           |
|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| $20\text{\$}(\text{Hourly pay})/47 \text{ Packets/hour} = 0.43\text{\$/Packet}$ | $20\text{\$}(\text{Hourly pay})/58 \text{ Packets/hour} = 0.34\text{\$/Packet}$ |

**Total Savings Per Bag** =  $0.43\text{\$}(\text{Cost Before}) - 0.34 (\text{Cost After}) = 0.09\text{\$/Packet}$

**Weekly Savings** =  $0.09\text{\$} * 16,922(\text{packets/week after pilot}) = 1,523 \text{\$/Week}$

**Monthly Savings:**  $1,523 \text{\$/week} * 4 = 6,092\text{\$/Month}$

**Yearly Savings:**  $6,092\text{\$/Month} * 12 = 73,104\text{\$/Year}$

After implementing the new process improvements, overall productivity increased from 47 packets per person per hour to 58 packets per person per hour. This improvement directly reduced the cost per packet from \$0.43 to \$0.34, assuming a standard hourly pay of \$20. While the change per unit may seem small, the impact at scale is significant — resulting in a savings of \$0.09 per packet. Based on the post-pilot weekly production volume of 16,922 packets, this translates to a cost savings of \$1,523 per week, or \$6,092 per month. Over the course of a year, the total projected savings amounted to \$73,104, demonstrating the long-term financial value of the operational improvements made during this project.

## RECOMMENDATIONS FROM IMPROVE PHASE

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Based on our in-depth analysis, we propose the following high-impact recommendations to improve packaging efficiency at Ayrloom. These solutions are rooted in our findings from time studies, spaghetti diagrams, and task audits.

### 1. Task Reallocation Based on Performance

- Assign faster workers (e.g., Morgan, Ana) to core cleaning and packing stations.
- Redirect slower but consistent workers (e.g., Amy, Kristen) to sticker application, boxing, and supportive logistics roles.
- Rotate moderately performing workers across tasks to build flexibility without slowing core lines.

### 2. Implement SOPs and Visual Aids for Cleaning and Packing

- Develop standardized steps with visuals for cleaning different gummy types (regular vs. sour).
- Introduce visual checklists at each station to ensure consistency in method and timing.
- Train new workers using shadowing + checklist walk-through.

### 3. Introduce Dedicated Runner Role

- Assign one support worker per shift to manage material logistics (stickers, boxes, gloves, bins).
- The runner will prevent motion waste and allow cleaners/packers to stay focused on core tasks.

### 4. Redesign Floor Layout to Support Lean Flow

- Implement a linear or U-shaped workflow: cleaning → packing → sealing → boxing.
- Centralize supplies and create isolated stations to eliminate crisscrossing paths.
- Add visual controls such as taped zones and labeled racks.

**5. Use Time Studies to Benchmark and Motivate**

- Display average times per task and per worker as internal metrics.
- Set achievable targets for each role and reward performance with verbal recognition or small incentives.

**6. Utilize Automatic Sealer to Reduce Bottlenecks**

- Train 2–3 employees on consistent use of the automatic sealer.
- Use automatic sealing as the default for high-volume batches, reserving manual sealing only for rework or small runs.

These six recommendations are practical, low-cost, and highly impactful. They are designed not only to improve throughput, but also to boost worker morale, reduce rework, and build a foundation for sustained scalability.

**1. Task Reallocation Based on Performance**

**2. Implement SOPs and Visual Aids for Cleaning**

**3. Introduce Dedicated Runner Role**

**4. Redesign Floor Layout to Reduce Motion Waste**

**5. Use Time Studies to Benchmark and Motivate**

**6. Utilize Automatic Sealer to Standardize Output**

# CONTROL PHASE

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## DEFINED ROLES IN THE OPTIMIZED PACKAGING PROCESS

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As part of the Control Phase, we clearly defined the responsibilities of each role in the updated gummy packaging process to ensure consistent execution, reduce idle time, and improve output quality. The introduction of the runner system and role-based task allocation required setting role clarity as a foundational control measure.

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### CLEANER

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#### **Primary Responsibility:**

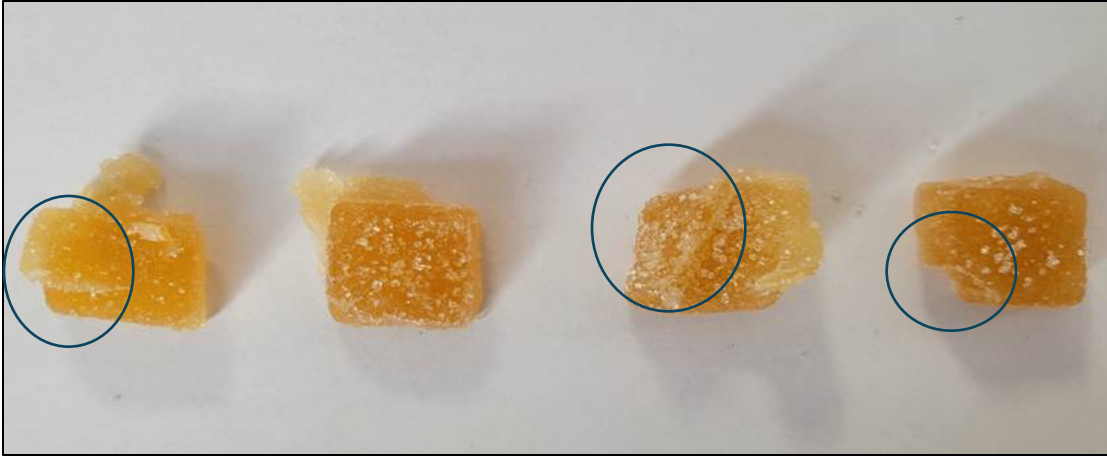
To clean raw gummies efficiently and perform visual quality inspection to remove defective pieces before forwarding to the packing stage.

#### **Key Responsibilities:**

- Receive boxes of raw gummies delivered by the runner.
- Clean gummies thoroughly following hygiene and workstation standards.
- Inspect each gummy visually and separate out defective pieces such as:
  - Broken gummies
  - Half-formed or melted gummies
- Place all defective gummies in a separate bag.
- Notify the runner once ready for the next batch of raw gummies.

#### **Expected Outcome:**

Delivery of clean, quality-approved gummies to the packing station with minimized movement and consistent visual standards.



Example of a Bad Gummy



Example of a Good Gummy

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### PACKER

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#### **Primary Responsibility:**

To pack cleaned gummies into designated packaging with speed and accuracy

#### **Key Responsibilities:**

- Receive cleaned gummies from the cleaner or assigned area.
- Count and pack exactly 10 gummies into each bag.
- Seal the bag using a Ziplock closure.
- Alert runner for additional packaging supplies when running low.

#### **Expected Outcome:**

Consistent and accurate packing of 10 gummies per bag, with properly sealed zip locks.

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## RUNNER

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### **Primary Responsibility:**

To support continuous workflow by delivering all necessary materials to cleaners and packers, eliminating the need for workers to leave their stations. The runner also handles the stickering process and ensures that all bins and materials are routed correctly as instructed by the team leader.

### **Key Responsibilities:**

- Transport raw gummy boxes to cleaners (in the order directed).
- Deliver cleaned gummies to packers in the correct sequence (in the order directed).
- Deliver packaging materials (e.g., empty packets) to packers.
- Perform stickering by attaching the correct batch number to each packet.
- Ensure the correct packet type is used for each gummy flavor and type.
- Monitor all workstations and proactively respond to supply needs
- Double-check and make sure the right supplies are delivered to the appropriate station

### **Expected Outcome:**

An uninterrupted and efficient production flow with minimal delays or errors. The runner plays a key quality gatekeeping role by ensuring the right materials reach the right people, with accurate stickering and attention to flavor-specific packaging.

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## TEAM LEAD (SUPERVISORY ROLE)

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### **Primary Responsibility:**

To oversee the shift, assign roles based on employee strengths, ensure smooth operations, and handle documentation and performance tracking related to the optimised process.

**Key Responsibilities:**

- Assign workers to cleaning or packing roles based on their speed, accuracy, and strengths identified during the Measure Phase.
- Monitor overall productivity, individual performance, and the efficiency of the runner system.
- Serve as the first point of contact for resolving workflow issues, delays, or imbalances.
- Track key performance indicators (KPIs), such as boxes packed per day.
- Communicate any process changes or important updates to the team.

**Expected Outcome:**

A well-balanced team with sustained productivity, accurate records of operations, and quick issue resolution, supporting the long-term success of the implemented improvements.

## PERFORMANCE MONITORING SYSTEM

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Due to limited supervisory bandwidth, we implemented a self-reporting time tracking system using Excel. This enables each employee (cleaner or packer) to log their own productivity in a structured format without requiring full-time observation.

**1. Manual Time Tracking Sheets**

Each employee is assigned an individual Excel sheet where they record:

- **Date**
- **Bin Number**
- **Time In** (when they start working on the bin)
- **Time Out** (when they finish cleaning or packing the bin)

|               |        |            |          |
|---------------|--------|------------|----------|
| Employee Name |        |            |          |
|               |        |            |          |
| Date          | Bin No | Start time | End time |
|               |        |            |          |
|               |        |            |          |
|               |        |            |          |
|               |        |            |          |
|               |        |            |          |

This method allows us to:

- Track how long each employee spends on each bin
- Identify performance variations across team members
- Analyze average cleaning and packing times per bin
- Maintain historical performance records for each employee

## 2. Psychological Accountability

Beyond just tracking time, this system creates a mental accountability factor. Since employees know their performance is being recorded and can be reviewed:

- They are more likely to remain focused and efficient
- It reduces idle time and encourages timely completion of tasks
- It allows the team lead to address consistently slower performance with data-backed insights

## AUTOMATIC ROLE ALLOCATION SYSTEM

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In the beginning, the team was assigning cleaning and packing roles manually. While that worked temporarily, it took a lot of thinking and time to figure out the right combinations. Also, since people improve over time, the same roles can't be assigned forever.

Even the previous work allocation was based on the data we (the project team) collected during our earlier visits. But we knew we wouldn't always be there to help decide.

So, we wanted to build something that the team lead could easily use every week without overthinking.

## **How the Process Works**

At the end of each week, the team lead:

- Collects the manual time tracking sheets from all employees.
- Randomly picks 10 bin records for each person from their cleaning and packing logs.
- Enters that data into an Excel sheet.

Since there are 8 employees, the team lead enters data into 8 individual sheets — one for each person. Each sheet has separate sections for cleaning and packing data.

Once that's done, all the individual averages from these sheets automatically feed into a master sheet, where all further calculations happen. This master sheet performs the following:

### **Step 1: The sheet does three things automatically:**

1. Calculates the average time each person takes to clean or pack one bin.
2. Ranks employees based on who is the fastest (lower time = better rank).
3. Converts the time into output/hour, i.e., how many bins each person can handle in one hour.

| CLEANING |          |          |            |      |            |               |
|----------|----------|----------|------------|------|------------|---------------|
|          | Time in  | Time out | Difference | Rank | Time taken | Bins per hour |
| Person 1 | 12:05 PM | 1:09 PM  | 1:04       | 3    | 1.07       | 0.94          |
| Person 2 | 2:35 AM  | 3:45 AM  | 1:10       | 4    | 1.17       | 0.86          |
| Person 3 | 12:00 PM | 1:45 PM  | 1:45       | 8    | 1.75       | 0.57          |
| Person 4 | 3:52 PM  | 5:12 PM  | 1:20       | 6    | 1.33       | 0.75          |
| Person 5 | 1:52 AM  | 3:14 AM  | 1:22       | 7    | 1.37       | 0.73          |
| Person 6 | 5:56 PM  | 7:12 PM  | 1:16       | 5    | 1.27       | 0.79          |
| Person 7 | 4:52 PM  | 5:42 PM  | 0:50       | 1    | 0.83       | 1.20          |
| Person 8 | 7:32 PM  | 8:35 PM  | 1:03       | 2    | 1.05       | 0.95          |

Cleaning Time recorded in Excel with Performance indicators and Rank

| PACKING  |          |          |            |      |            |               |
|----------|----------|----------|------------|------|------------|---------------|
|          | Time In  | Time out | Difference | Rank | Time taken | Bins per hour |
| Person 1 | 1:10 PM  | 1:50 PM  | 0:40       | 1    | 0.67       | 1.50          |
| Person 2 | 5:35 AM  | 6:45 AM  | 1:10       | 7    | 1.17       | 0.86          |
| Person 3 | 3:00 PM  | 5:45 PM  | 2:45       | 8    | 2.75       | 0.36          |
| Person 4 | 6:52 PM  | 7:55 PM  | 1:03       | 5    | 1.05       | 0.95          |
| Person 5 | 7:50 PM  | 8:55 PM  | 1:05       | 6    | 1.08       | 0.92          |
| Person 6 | 2:52 AM  | 3:45 AM  | 0:53       | 3    | 0.88       | 1.13          |
| Person 7 | 6:30 AM  | 7:20 AM  | 0:50       | 2    | 0.83       | 1.20          |
| Person 8 | 10:19 AM | 11:15 AM | 0:56       | 4    | 0.93       | 1.07          |

Packing Time recorded in Excel with Performance indicators and Rank

The Metrics that are marked in the Images are calculated automatically

## Step 2: What Solver Does

Once we know each person's speed, we run Solver in Excel to figure out the best possible combination of cleaners and packers for that week.

We gave Solver the following conditions:

- Every person should be assigned to only one task — either cleaning or packing.

- The difference between total bins cleaned and total bins packed per hour is set to be less than or equal to 0.3(Which can be changed as needed), so there's no backup or waiting around.
- The Cleaning output should be greater than packing output (to incentivize cleaners with sealing)
- The total number of bins processed per hour (cleaning + packing) should be as high as possible — that's the main goal.

Solver tries out all possible combinations and shows the one that gives the best balance and maximum output.

Set Objective:

To: ☒ Max ☐ Min ☐ Value Of:

By Changing Variable Cells:

Subject to the Constraints:

- 
- 
- 
- 

☒ Make Unconstrained Variables Non-Negative

Select a Solving Method:

Solving Method

Select the GRG Nonlinear engine for Solver Problems that are smooth nonlinear. Select the LP Simplex engine for linear Solver Problems, and select the Evolutionary engine for Solver problems that are non-smooth.

Buttons: Add, Change, Delete, Reset All, Load/Save, Options

### Solver Parameters

### Step 3: Output Example

From the Solver result below:

- 5 employees were assigned to cleaning, and 3 employees were assigned to packing
- **Cleaning Output:** 3.86 bins/hour
- **Packing Output:** 3.83 bins/hour

- **Difference:** 0.03 bins/hour (within allowed limit)
- **Total Combined Output:** 7.81 bins/hour

| Person         | Cleaner | Packer                | Assigned? |
|----------------|---------|-----------------------|-----------|
| Person 1       | 1       | 0                     | 1         |
| Person 2       | 1       | 0                     | 1         |
| Person 3       | 1       | 0                     | 1         |
| Person 4       | 0       | 1                     | 1         |
| Person 5       | 0       | 1                     | 1         |
| Person 6       | 0       | 1                     | 1         |
| Person 7       | 1       | 0                     | 1         |
| Person 8       | 0       | 1                     | 1         |
| on c\p         | 4       | 4                     |           |
| Cleaning       |         | 3.56607 bins per hour |           |
| Packing        |         | 3.47786 bins per hour |           |
| Difference     |         | 0.08821 bins per hour |           |
| Total bins C+P |         | 7.04393 bins per hour |           |

#### Step 4: Output Visualization – Most Efficient Combination

To make it easier for the team lead to review Solver results at a glance, we created a clean summary table that shows the **recommended combination** of cleaners and packers, their individual outputs, and the total output per hour.

| Most Efficient combination |      |                                         |          |      |               |
|----------------------------|------|-----------------------------------------|----------|------|---------------|
| CLEANING                   |      |                                         | PACKING  |      |               |
| Bins Per Hour              | Rank | Cleaners                                | Packers  | Rank | Bins per Hour |
| 0.94                       | 2    | Person 1                                | Person 4 | 4    | 0.95          |
| 0.86                       | 3    | Person 2                                | Person 5 | 5    | 0.92          |
| 0.57                       | 7    | Person 3                                | Person 6 | 7    | 0.53          |
| 1.20                       | 1    | Person 7                                | Person 8 | 3    | 1.07          |
| 0.00                       | 0    |                                         |          | 0    | 0.00          |
| 3.57                       |      |                                         |          |      | 3.48          |
| TOTAL OUT PUT PER HOUR =   |      | 3.57(Cleaned bins) + 3.48 (Packed Bins) |          |      |               |

#### What This Table Shows:

- The left side lists employees assigned to **cleaning**, along with their bins/hour and rank.
- The right-side lists those assigned to **packing**, also showing their speed and rank.
- The bottom row shows the **total combined output**, in this case:
  - **Cleaning Output:** 3.86 bins/hour

- **Packing Output:** 3.83 bins/hour
- **Total Output:** 7.69 bins/hour

### Step 5 Sealer allocation

This output table helps the team lead & the Management to:

- Understand the result of the Solver without reading through formulas.
- Explain why each person was assigned to a specific task.
- Compare this week's combination to previous ones and track improvements.

## REACTION PLAN

Even with improvements in place, things can still go wrong during daily operations. To handle such situations quickly and avoid slowdowns, we created a simple **reaction plan**. This outlines the most common issues that might happen in the cleaning and packing process, the likely reasons behind them, and what immediate action should be taken.

This plan helps the team lead respond quickly when performance drops, someone misses a step, or when work becomes unbalanced.

| What Might Go Wrong                                                   | Likely Cause                                                  | Immediate Response                                                     | Responsible Person   |
|-----------------------------------------------------------------------|---------------------------------------------------------------|------------------------------------------------------------------------|----------------------|
| <b>Drop in boxes packed per day</b>                                   | Incorrect role allocation, slower performance                 | Review the time tracking sheet, adjust roles the next day using Solver | Team Lead            |
| <b>The worker is idle, waiting for the next bin/lack of resources</b> | The runner was delayed or not rotating efficiently            | Team Lead temporarily steps in as backup runner                        | Team Lead            |
| <b>Long time to clean/pack a single bin</b>                           | The worker is distracted, tired, or unclear about the process | Talk to the worker, check if training or a role change is needed       | Supervisor           |
| <b>Employees are not filling out time sheets properly</b>             | Forgot or unwilling                                           | Remind during the shift, and follow up individually                    | Team Lead/Supervisor |

## BEFORE-AND-AFTER COMPARISON

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The table below highlights the key differences observed before and after implementing the new runner system and optimized role allocation. The goal was to reduce unproductive time, improve task balance, and increase hourly output using data-backed decisions. Each metric shown reflects an area where process changes had a direct operational impact — from how tasks were assigned, to how much movement was required, to how much each worker could produce in a shift.

| Metric                        | Before                           | After                                 | Improvement                 |
|-------------------------------|----------------------------------|---------------------------------------|-----------------------------|
| <b>Task allocation</b>        | Everyone doing the same task     | Split roles based on individual speed | Better balance and output   |
| <b>Worker movement</b>        | High (each worker fetched boxes) | Zero (runner delivers to stations)    | Reduced idle time & fatigue |
| <b>Output per worker/hour</b> | ~47 boxes                        | ~58 boxes                             | 23.5%                       |

## FUTURE SCOPE

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### REDUCING CLEANING TIME: A KEY AREA FOR FURTHER PRODUCTIVITY GAINS

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While the improvements in role allocation and packing efficiency have led to a significant boost in productivity, **cleaning remains a major bottleneck**, making it the most labor-intensive part of the process.

Upon deeper analysis, we found that **the root cause originates in the kitchen**. During production, the gummy slurry is not evenly spread across the molds, leading to excess material sticking to the gummies. This residue is what makes the cleaning process so time-consuming.

To further improve productivity, we believe the solution lies in a **two-pronged approach**:

1. **Avoiding the issue at the source** — by improving how slurry is applied in the kitchen, possibly using a **dual-scrapers system** to prevent excess buildup.
2. **Finding a better way to clean** — either through tooling, automation, or a simplified manual process that reduces the time required.

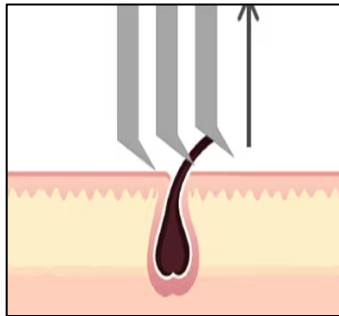
Addressing the problem from both ends — **prevention in the kitchen and optimization in cleaning** — will be essential for unlocking the next level of efficiency.

### DUAL-SCRAPER SYSTEM FOR CLEANER GUMMIES

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One of the main reasons gummies require manual cleaning in the post-production phase is due to **excess gummy material sticking around the edges**. This happens because the slurry from the reactor is directly poured into the mold and spread with a basic scraper. However, this scraper often leaves gummy residue or overflow attached to the sides of the molded pieces, making the cleaning process more labor-intensive.

To address this, we propose the development of a **dual-scrapers system**, inspired by how **razor blades** work — where the first blade spreads and the next ones follow through to clean the remaining bits.



How a Razor works

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### PROPOSED MECHANISM

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- **First Scraper:** Spreads the slurry evenly across the mold, as it currently does.
- **Second Scraper:** Positioned slightly behind the first one, this blade will gently shave off or collect any extra slurry that wasn't properly flattened — preventing buildup on the gummies.

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### EXPECTED BENEFITS

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- Reduce the need for intensive manual gummy cleaning
- Lower labor time and cost in the cleaning stage
- Produce more consistent and clean gummy shapes
- Improve overall product appearance and quality at the source

This mechanical upgrade can potentially eliminate the root cause of excess gummy buildup and streamline the process further. Prototyping and small-scale trials of the dual-scrapers concept could be the next step.

# CHANGE MANAGEMENT INSIGHTS FROM THE CONTROL PHASE OF LEAN SIX SIGMA PROJECT

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During our Control Phase meeting, we made a significant breakthrough in understanding why our Lean Six Sigma implementation required more than just technical adjustments to succeed. We identified that sustainable improvement demands a synergistic approach combining process optimization with leadership development, team dynamics management, and organizational structure. Our analysis revealed that resistance to change, goal visibility issues, leadership development needs, and performance tracking were interconnected challenges that needed to be addressed holistically. The packaging team's performance wasn't just a matter of production metrics—it required nurturing a culture where change is embraced, goals are transparent, leadership is empowered, and progress is visible to all stakeholders. This comprehensive approach has proven essential for translating technical improvements into lasting operational excellence and team engagement.

Based on these findings, we have conducted additional research on how to address these issues successfully, and we focused on both Team Leadership level, as well as Senior Leadership. The four main topics are as follows:

1. Change management
2. Work rotation and role distribution
3. New leadership, setup for success
4. Goal visibility – The dashboard

## CHANGE MANAGEMENT

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### INSIGHTS FOR IMMEDIATE LEADERSHIP

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#### **Prepare Your Team for Change**

For an organization to successfully pursue and implement change, it must be prepared both logistically and culturally. Before delving into logistics, cultural preparation must first take place to achieve the best business outcome.

For your gummy-packing team:

- Hold a dedicated meeting to explain why processes are changing (increased efficiency, market demands, quality control, etc.)
- Create a safe space for workers to express concerns about new roles and responsibilities
- Acknowledge the expertise they've developed in their previous roles

#### **Develop a Clear Vision and Plan**

Once the organization is ready to embrace change, managers must develop a thorough, realistic, and strategic plan for bringing it about.

For your gummy-packing team:

- Create visual aids showing the new workflow process
- Develop clear role descriptions for each team member
- Set realistic production targets that account for adjustment time
- Establish milestones for the transition period (1 week, 2 weeks, 1 month)

## **Implementation Strategy**

During the implementation process, change managers must be focused on empowering their employees to take the necessary steps to achieve the goals of the initiative and celebrate any short-term wins.

For your gummy-packing team:

- Schedule regular brief check-ins (5-10 minutes) at shift beginnings/ends
- Create a visual tracker for production metrics to show progress (Dashboard)
- Celebrate when the team hits new milestones (pizza lunch, recognition board)

## **Embed Changes in Daily Operations**

Once the change initiative has been completed, change managers must prevent reversion to the prior state or status quo. This is particularly important for organizational change related to business processes such as workflows.

For your gummy-packing team:

- Update all visual aids and procedure documents
- Remove old equipment or rearrange workspace to support new processes
- Create simple checklists for new procedures
- Recognize and reward adherence to new processes

## **Review Progress and Make Adjustments**

Conducting analysis and review can help business leaders understand whether a change initiative was a success, failure, or mixed result.

For your gummy-packing team:

- Track production rates, quality metrics, and worker satisfaction

- Schedule a review meeting 30 days after implementation
- Ask for improvement suggestions from the team
- Be willing to make minor adjustments based on worker feedback

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## INSIGHTS FOR SENIOR LEADERSHIP (ADMINISTRATIVE AND OPERATIONAL)

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### **Understand Change Management Fundamentals**

Change management is a set of ideas, strategies, and skills that can be applied to engage change effectively. These may be applied in planning for change, implementing change, and supporting continuous improvement following change.

For your organization:

- Recognize this gummy-packing process change as an opportunity to build change management capacity
- Support immediate supervisors with resources and training
- Ensure alignment between this process change and broader organizational goals

### **Assess Readiness for Change**

A guideline for assessing the likely success of a proposed change requires evaluating three key elements: the leadership capacity and attention span for driving the change, the business need for the change, and the energy of affected people for and towards the change.

For your organization:

- Ensure the immediate supervisor has adequate time and resources to manage this change
- Clearly articulate how this process change connects to market needs/business goals
- Gauge whether additional support is needed based on team morale

## **Communication Strategy**

For a communication plan, a leader should communicate about each step of the change "seven different times and in seven different ways" to encourage support for the change and help ensure its effectiveness.

For your organization:

- Support multi-channel communication about the changes (meetings, posters, digital updates)
- Ensure consistent messaging across different levels of leadership
- Provide a clear rationale for why this change benefits both the company and workers

## **Support Continuous Improvement**

Change management methods may be applied in supporting continuous improvement following change.

For your organization:

- Schedule regular reviews of the new process (30, 60, 90 days)
- Consider what learnings from this change can be applied to other departments
- Develop metrics to track both productivity and worker satisfaction with new roles

## **Stakeholder Management**

At the outset of a change process, a leader should meet with each major stakeholder group. Combine active sharing of the benefits and tradeoffs of the change with active listening to stakeholder concerns.

For your organization:

- Ensure quality control, sales, and warehouse teams understand how the gummy-packing changes affect them
- Consider whether customer communication is needed if packaging appearance will change
- Meet with suppliers if material requirements will change

## ROTATING SCHEDULE TIPS FOR GUMMY PACKING TEAM MANAGEMENT

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### BENEFITS OF IMPLEMENTING A ROTATING SCHEDULE

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Rotating schedules can be tremendously effective when implemented correctly. They provide employees with a complete picture of operations at different times and strengthen your team as employees get the chance to work with different teammates. This will be particularly valuable for your gummy-packing operation where understanding the full process can improve overall efficiency.

### SPECIFIC ROTATION RECOMMENDATIONS FOR YOUR TEAM

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#### **Role Rotation Structure**

Given your team composition (4 Cleaners, 4 Packers, 1 Runner), consider implementing:

A rotating schedule where your employees work different roles across different scheduling cycles. For example, in a given two-week scheduling cycle, an employee might work different roles during different weeks.

#### **Recommended Structure:**

- Establish a 3-week rotation cycle
- Week 1: Original roles (4 Cleaning, 4 Packing, 1 Runner)
- Week 2: Rotate 2 Cleaners to Packing, 2 Packers to Cleaning, and 1 Packer to Runner

- Week 3: Return to original positions but with different teammate pairings

This ensures everyone maintains proficiency in their primary role while gaining experience in other positions.

### **Training and Transition Plan**

Creating a rotating schedule requires a step-by-step approach. First, determine what skills you need for each shift and how many employees you need to provide full coverage. Consider skill sets that only some people are trained for.

For your team:

- Create detailed role descriptions for each position (Cleaner, Packer, Runner)
- Implement a "buddy system" where experienced workers in each role train newcomers
- Allow for a 1-2 day learning curve when workers transition to a new role
- Create visual aids showing specific techniques for each position

### **Team Composition Strategy**

By creating teams, you'll be able to rotate them through different roles. Creating teams is a skill in itself, as you might want to group together people with complementary skills, differing levels of seniority, and different responsibilities.

For your gummy operation:

- Ensure each rotation includes a mix of experienced and newer employees
- Consider pairing faster and slower workers to balance production flow
- Match employees with complementary working styles (detail-oriented with efficiency-focused)
- Keep at least one highly experienced person in each role during transitions

## **Schedule Communication and Accessibility**

Your rotating schedule needs to be easily accessible and editable in order to account for irregularities and last-minute changes. Having schedules easily available will make it easier to account for unexpected changes.

For your team:

- Post the rotation schedule at least two weeks in advance
- Create both a physical posted schedule and a digital version
- Use color coding to make the schedule visually clear (different colors for different roles)
- Include a simple process for handling emergency swaps or absences

## **Performance Tracking and Adjustments**

Try out different combinations of rotation schedules to balance your employee and business needs. It'll be helpful to track attendance so you can fill open shifts promptly when no-shows or emergencies occur.

For your operation:

- Track productivity metrics during each rotation phase
- Note which employee combinations work best in which roles
- Hold brief weekly meetings to discuss rotation challenges and successes
- Be willing to modify the rotation system based on real performance data

## **Addressing Potential Challenges**

Employees may not like the idea of rotating shift work. Some people like the consistency of fixed roles and predictable work responsibilities. If a business is to deploy a

rotating schedule, it's essential to give employees notice of their shifts far enough in advance.

To mitigate resistance:

- Explain the benefits of role rotation clearly (skill development, reduced monotony)
- Implement a gradual transition to the rotation system
- Create incentives for learning new roles (recognition, small bonuses)
- Allow employees who strongly prefer one role to spend more time there while still gaining exposure to others

### **Equipment and Workspace Considerations**

Since your team involves specialized roles with different tasks:

- Ensure all workstations are standardized for easy transition between workers
- Create simple checklists for each role to guide newer workers
- Label all tools and supplies clearly
- Design the physical workspace to allow for easy movement between stations

### **Special Considerations for the Runner/Supplier Role**

Since you only have one Runner position:

- This is the most critical role to rotate, as it provides understanding of the entire process
- Have all team members experience this role eventually
- Create clear training for this position, as it coordinates the work of others
- Consider having the most experienced employee shadow new Runners during their first rotation

## **Review and Optimization**

Give as much notice to your employees as possible before you set your rotating schedule, so employees get plenty of time to adapt to it, and your rotating schedule system is ultimately successful.

For ongoing improvement:

- Schedule a full team review after completing the first 3-week rotation cycle
- Ask for specific feedback about each role transition
- Be willing to adjust rotation frequency based on team input
- Document "best practices" that emerge from each rotation

## **GUIDE FOR FIRST-TIME TEAM LEADER SUCCESS IN THE PACKAGING DEPARTMENT**

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### **DAILY TASK MANAGEMENT FOR THE NEW TEAM LEADER**

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#### **Establish a Consistent Daily Routine**

As a new leader of the packaging team, develop a structured daily routine that includes:

- Begin each day with a brief team huddle (5-10 minutes) to set priorities and address any concerns
- Schedule time each day for one-on-one check-ins with different team members on a rotating basis
- Block specific hours for administrative tasks like reviewing production metrics
- End each day with a quick review of what was accomplished and what needs attention tomorrow

#### **Focus on Communication Excellence**

Communication is your most valuable leadership tool:

- Practice active listening when team members share ideas or concerns
- Become skilled at giving constructive feedback that is specific and actionable
- Document important discussions and decisions to maintain clarity
- Learn to tailor your communication style to different team members' preferences

### **Develop Fair Decision-Making Processes**

Build trust through transparent decision-making:

- Establish clear criteria for decisions that affect the team
- When appropriate, involve team members in decision-making processes
- Explain the reasoning behind difficult decisions
- Be consistent in how you apply rules and policies

### **Manage Your Time Effectively**

Protect your ability to lead by managing time wisely:

- Distinguish between urgent and important tasks
- Delegate appropriate responsibilities to capable team members
- Schedule uninterrupted focus time for planning and strategic thinking
- Learn to say no to tasks that don't align with team priorities

### **Build Team Culture Intentionally**

Shape the work environment through daily actions:

- Recognize and celebrate small wins and team member contributions
- Model the behavior you expect from others
- Address conflicts promptly and fairly
- Create opportunities for team building during regular work hours

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## GUIDANCE FOR SENIOR LEADERSHIP SUPPORTING THE NEW TEAM LEADER

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### **Provide Structured Onboarding Support**

Help your new team leader transition successfully:

- Create a 30-60-90 day plan with clear expectations and milestones
- Schedule weekly check-ins during the first three months
- Connect the new leader with peers who can provide informal guidance
- Be clear about decision-making authority and when escalation is appropriate

### **Offer Meaningful Development Opportunities**

Invest in your new leader's growth:

- Provide access to relevant training or workshops on leadership fundamentals
- Assign a seasoned mentor from another department
- Encourage shadowing opportunities with experienced leaders
- Create safe spaces to practice new skills and receive feedback

### **Establish Clear Performance Expectations**

Set your new leader up for success with clarity:

- Define what success looks like in specific, measurable terms
- Balance attention to both technical outcomes and leadership behaviors
- Create a feedback loop that allows for regular course correction
- Recognize that different aspects of leadership develop at different rates

## **Create Safe Spaces for Learning**

Normalize the learning curve of leadership:

- Encourage the new leader to ask questions without fear of judgment
- Allow appropriate room for mistakes and learning
- Share your own leadership challenges and how you overcame them
- Celebrate progress and improvement, not just perfection

## **Provide Tools and Resources**

Equip your new leader with necessary support:

- Ensure access to relevant data and metrics for informed decision-making
- Offer tools for planning, scheduling, and team management
- Create access to HR support for navigating personnel challenges
- Build a library of leadership resources tailored to your organization

## PRODUCTIVITY DASHBOARD



The Gummy Packing Team Dashboard transforms abstract productivity goals into visible metrics that motivate the entire team. It displays current role assignments alongside daily, weekly, monthly, and yearly production data compared against the target of 60 packs per day. Through intuitive color-coded indicators (red for below target, green for meeting/exceeding), team members can instantly assess both individual and collective progress, creating shared accountability and fostering a results-oriented culture where everyone understands not just their roles but how their work contributes to overall success.

## CONCLUSIONS

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The Lean Six Sigma initiative undertaken at Ayrloom marks a significant milestone in operational excellence for Beak & Skiff's expanding cannabis edibles division. Confronted with a rapidly growing production demand, the project addressed the critical bottleneck in the manual gummy packaging process. By applying the structured DMAIC methodology, the team identified performance gaps rooted in inconsistent worker speeds, lack of standard operating procedures, inefficient task allocation, and excessive motion due to poor floor layout and material flow.

Through comprehensive time studies, observational data, and Lean Six Sigma tools such as Fishbone Diagrams, Pareto Analysis, and Spaghetti Diagrams, the team pinpointed cleaning inefficiencies as the primary constraint and recognized the need for a more systematic approach to task delegation. The implementation of a dedicated runner role, data-driven role assignments using Excel Solver, visual controls, and a restructured U-shaped layout significantly reduced idle time and unnecessary movement. A pilot test confirmed the effectiveness of these interventions, resulting in a 23.4% increase in productivity and projected annual savings of over \$43,000 in labor costs.

Beyond quantitative results, the project fostered a cultural shift toward accountability, transparency, and continuous improvement. Change management strategies, visual dashboards, rotating schedules, and leadership development initiatives were embedded to sustain gains and empower frontline workers and supervisors alike.

This project demonstrates how a modest, targeted intervention when driven by accurate data and team collaboration can produce scalable and sustainable results. It not only resolves existing process inefficiencies but also establishes a replicable model for future improvements across the organization. As Ayrloom continues to grow, this Lean Six Sigma framework will serve as a powerful tool for maintaining quality, efficiency, and employee engagement in an increasingly competitive market.

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